

ME5412 Robotics for Healthcare

Manipulation aids

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Outline of the lecture

- Target users of manipulation aids
- Low-tech manipulation aids, for general use
- Low-tech manipulation aids, for specific use
- High-tech manipulation aids, for specific use
- High-tech manipulation aids, for general use
- Assistive robotic manipulators (ARMs)

Target users of manipulation aids

- **Manipulation:** those activities that we normally accomplish using the upper extremities, particularly the fingers and hand
 - **Gross motor manipulation** (larger and more forceful movement and function): reaching, grasp/release, lifting, carrying, and coordination of movements such as pushing, pulling, throwing/catching, and turning
 - **Fine motor manipulation** (smaller and more delicate movements): pinch, point, and dexterity of finger movements
- The end goal of the integration of these manipulation components is a person's actions in daily and life activities:
 - handwriting, food preparation, eating, reaching, and using controls in the environment such as door handles, elevator buttons, and appliance buttons depend on manipulation of physical objects.
- **Target users** are individuals with upper-limb disabilities:
 - _____
 - _____

Name a target user for manipulation aids (from previous lecture)

✓ 0

Nobody has responded yet.

Hang tight! Responses are coming in.

Target users of manipulation aids

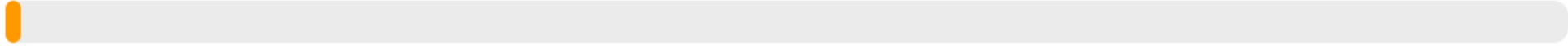
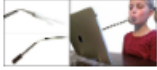
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 - handwriting, food preparation, eating, reaching, and using controls in the environment such as door handles, elevator buttons, and appliance buttons depend on manipulation of physical objects.
- **Target users** are individuals with upper-limb disabilities:
 - Quadriplegic individuals (spinal cord injuries)
 - Individuals with limited motor abilities, in terms of range of motion and/or strength

Classification of manipulation aids

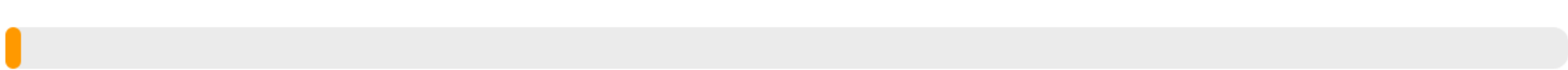
	Low-tech	High-tech
Specific tasks (<i>e.g., feeding</i>)		
General use		

Which are low-tech manipulation aids, for general use?

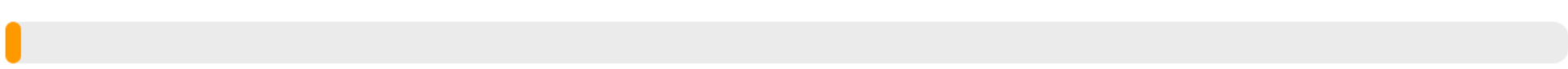
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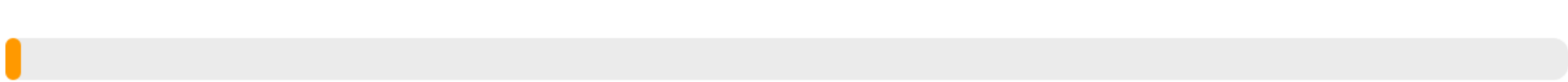
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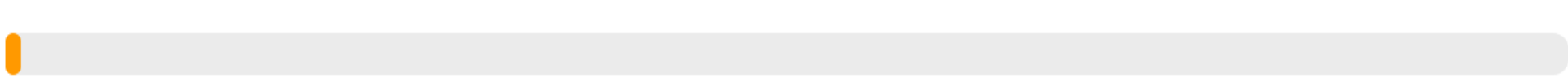
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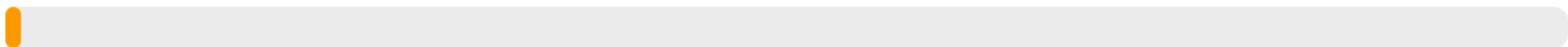
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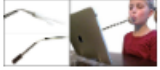
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Which are low-tech manipulation aids, for specific use?

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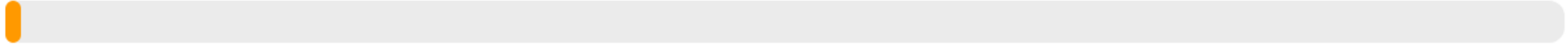
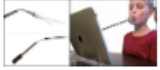
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Which are high-tech manipulation aids, for specific use?

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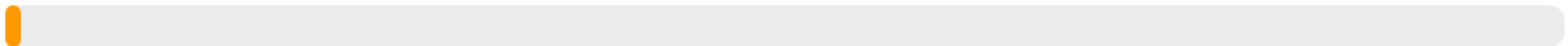
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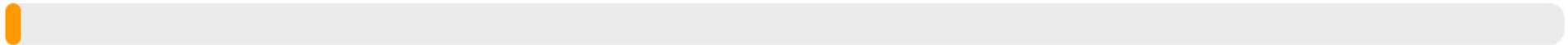
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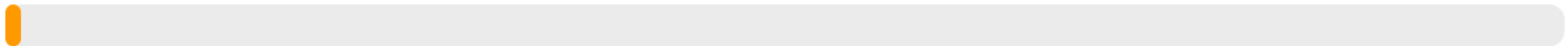
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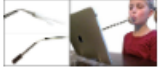
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Which are high-tech manipulation aids, for general use?

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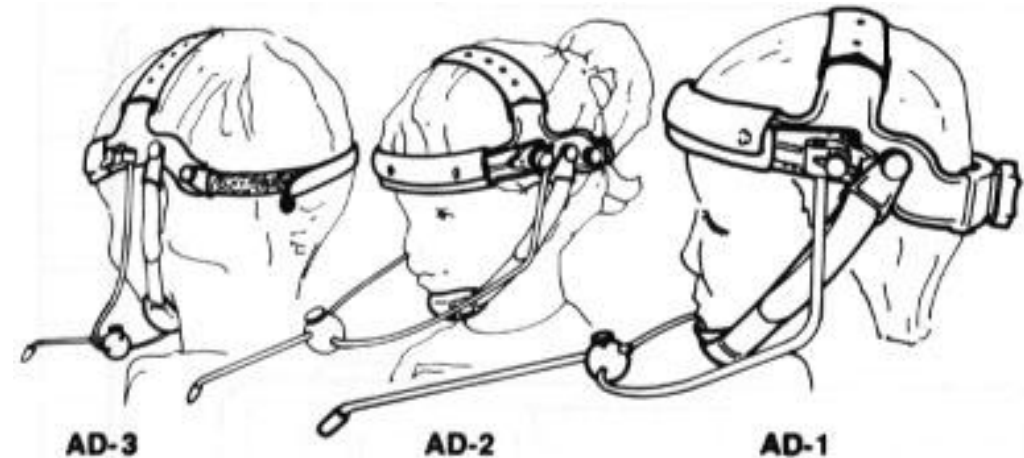
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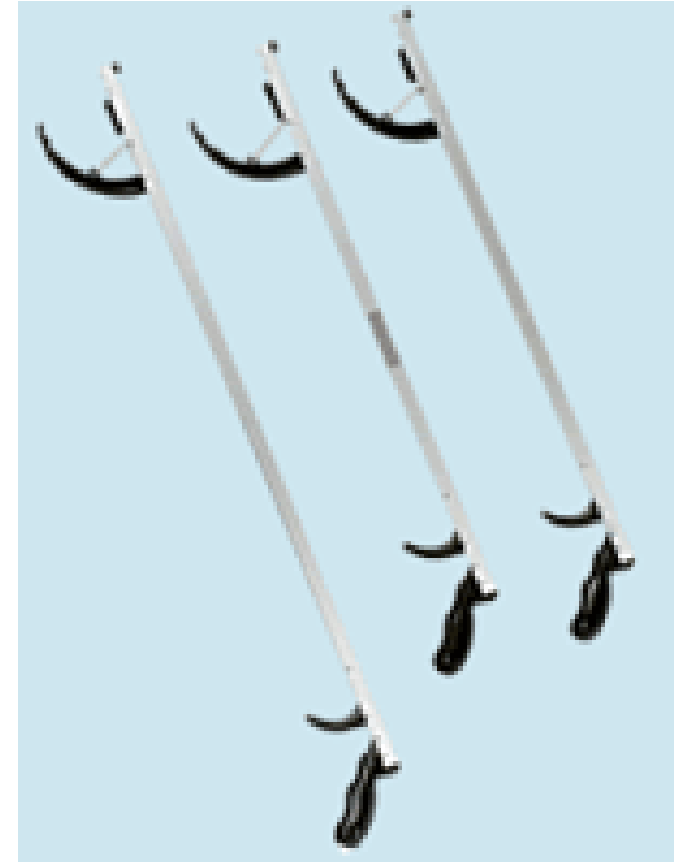
Low-tech manipulation aids, for general use

- A **mouth stick** is usually a wooden dowel with different attachments at one end and a molded piece at the other end that allows the person to hold the stick in mouth.
- A **head pointer** is similar in design, except the dowel is fixed to a headband that is worn around the forehead.
- Both are controlled by head movements.
- A pen or a pencil can also be attached to a mouth stick for writing.
- Additional attachments include a pincher that is opened or closed by tongue action and a suction cup end that can be used to grip objects (e.g., a page) by sucking on the end of the mouth stick.
- Mouth sticks or head pointers can also be used for such functions as typing and turning lights on and off.



Low-tech manipulation aids, for general use

- **Reachers** extend range of motion of
 - Wheelchair users
 - Individuals that have bending difficulties
- A **gripper** on one side and a **handle** with a lever to close the gripper
- Handle grasping may be:
 - With full hand
 - With all fingers
 - With the index finger
- Used to manipulate food (e.g., cans, packages), cooking utensils (e.g., pans, pots, plates, dishes), office objects (e.g., paper, books, magazines), and recreational or leisure objects (e.g., books, playing cards, eReaders).
- Activities such as dressing, hygiene and other self-care activities, and household activities—any activity that requires grasp and retention of an object that is outside of the individual's reach.
- **IMPORTANT:** because a reaching device extends the length of the arm, it also extends the length of the lever arm, resulting in increased force required to lift the object.



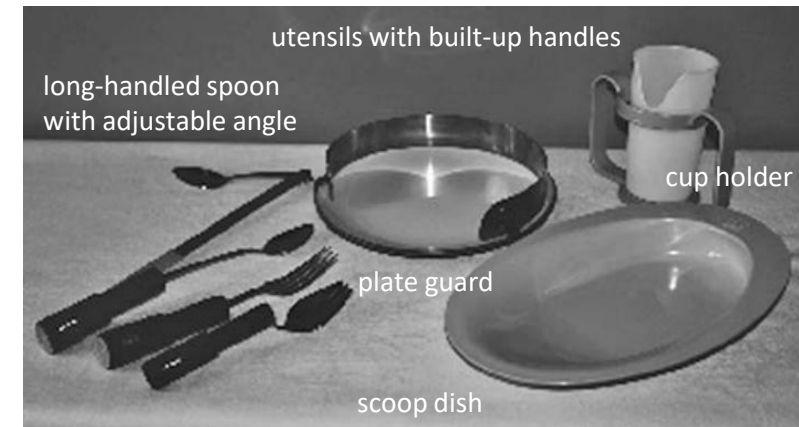
Low-tech manipulation aids, for specific use

- Designed for one or two tasks only
 - they serve those tasks very well
 - however, several are needed to meet the demands of self-care, work, and leisure.
- Often standard tools, adapted through:
 1. lengthening a handle or reducing the reach required
 2. modifying the handle of a utensil for easier grasping or manipulation
 3. converting two-handed tasks to one-handed ones (providing an alternative method of stabilizing the tools that are being used)
 4. amplifying the force that a consumer can generate with hands.

Low-tech manipulation aids, for specific use

Special-purpose adaptations of products include

- Enlarged grips for easier grasping,
- Cuffs that hold a utensil and circle the fingers,
- Angled handles for ease of scooping (for people with limited wrist movement),
- Swivel handles that allow the end to be oriented differently for different positions in space (e.g., on a table or near the mouth),
- Handles requiring limited grasp.



High-tech manipulation aids, for specific use

Page turners

- Two primary actions:
 1. separating the page to be turned from the other pages
 2. physically moving the page from one side to the other (forward or backward).
- Additional useful but not essential features include
 1. scanning several pages
 2. turning to a specific page
 3. locating a bookmark
 4. turning to that page
- Turning a single page is the most difficult task for page turners. Once the page to be turned is successfully isolated, the page turner must move it to the opposite side of the book or magazine.
- Page turners use several different mechanisms:
 - rubber-coated wheel to separate the pages, and then a rotating semicircular disk to push the separated page from one side to the other. As the disk rotates, the page is moved forward or backward, depending on the direction of rotation of the disk.
 - a vacuum to suck the top page away from the rest of the book and then turn the page.
- Advances in mainstream page turners have been driven by the need for musicians to have hands-free page turning.
- Typically, page turners use two switch inputs: forward and backward.

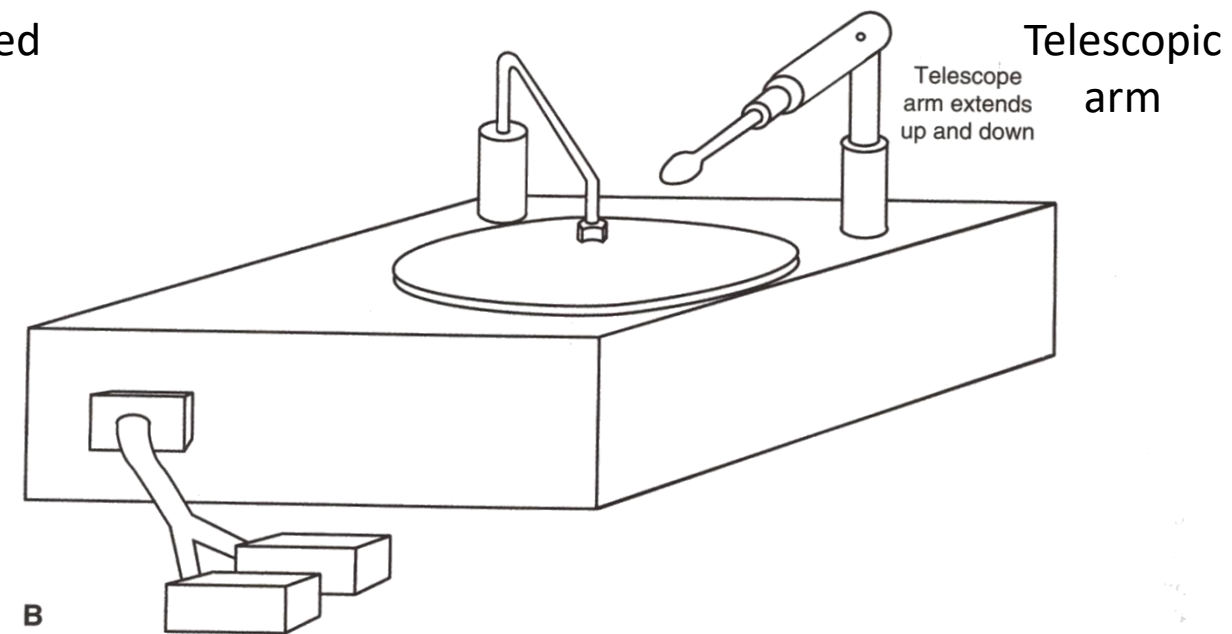
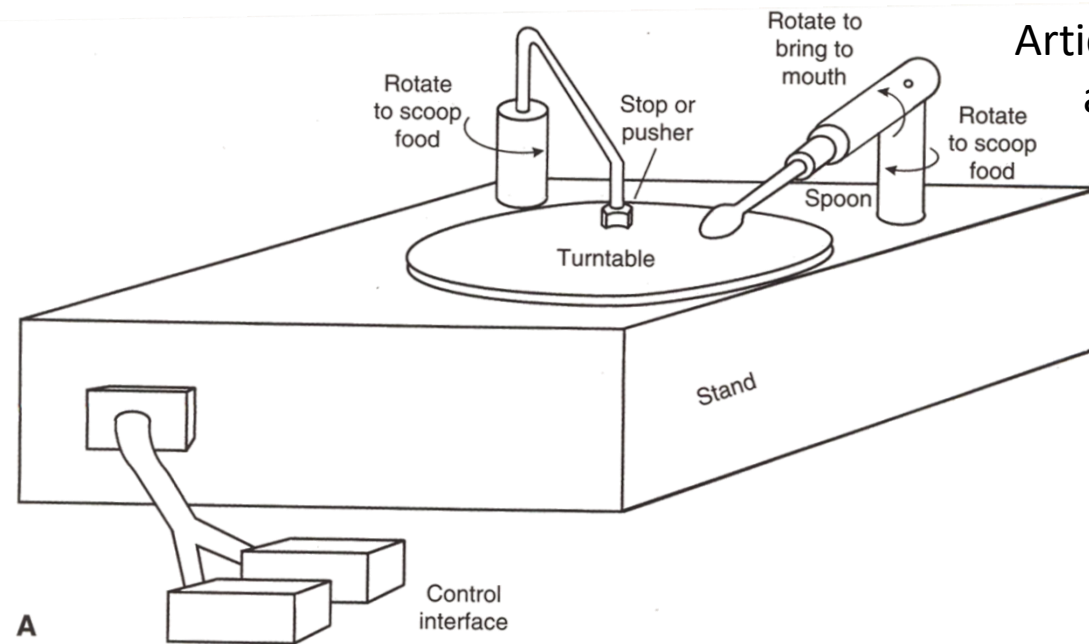


High-tech manipulation aids, for specific use

Feeders

Two functions:

- To locate the food
- To scoop the food



Food on turntable, the user stops turning when desired food is properly positioned

How many DOF are needed in a feeder?

✓ 0

1

2

3

4

5

6

> 6

SEE MORE 

OBI Robotic Feeder



<https://www.youtube.com/watch?v=s8gMf1BhX6o&t=1s>

Neater Eater by Neater Solutions Ltd, UK

2 DOF in the arm:

- Up/down
 - Forward/backward
- 1 DOF in the spoon
- Rotation



<https://www.neater.co.uk/neater-eater>

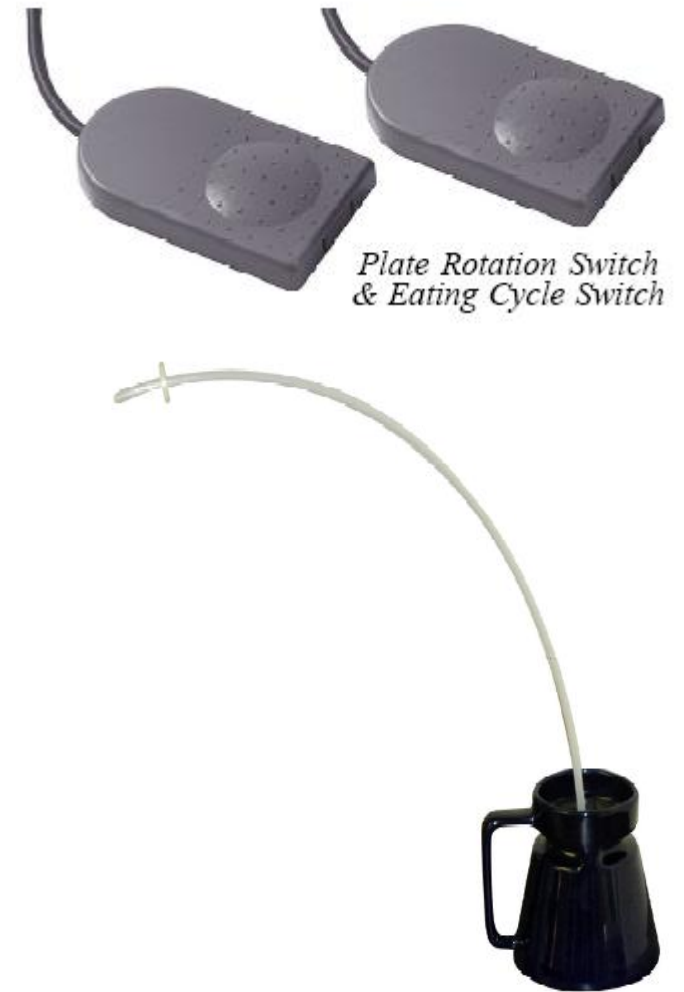
<https://youtu.be/Nm5XZG059Ew?si=6SQF7IGCjhjbAcB->

Neater Eater Control

4 control modes for the spoon:

1. One cycle: the spoon moves once
2. The spoon moves only if the switch is pressed
3. The spoon moves continuously
4. Single cycle: it starts when dish rotation is also activated

The cup is not grasped but the user drinks from the straw



Handy 1 by RehabRobotics, UK

- Programmable robotic arm
- 5 DOF



Weight: 25 Kg
Height: 99 cm ÷ 120cm
Width: 76 cm

How can the user control 5 DOF (with limited motor abilities)?

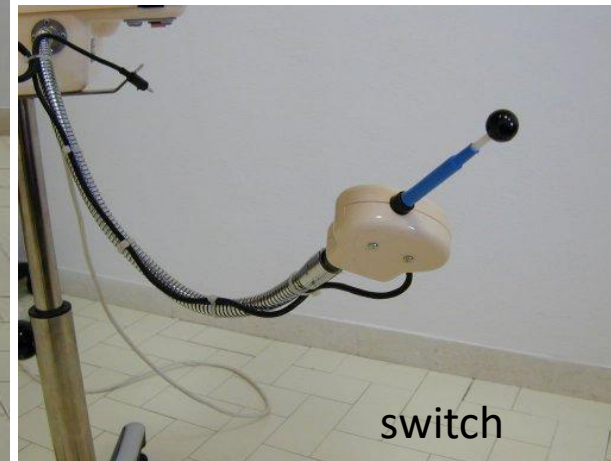


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Hang tight! Responses are coming in.

Handy 1 by RehabRobotics, UK

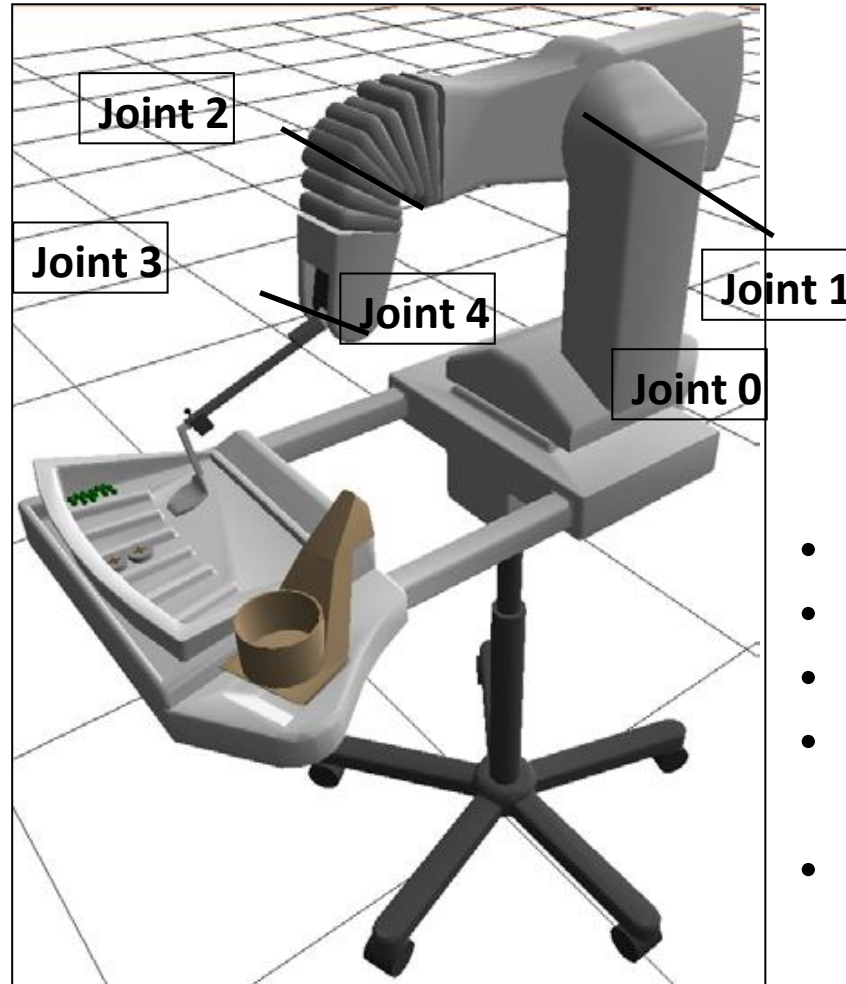
- Programmable robotic arm
- 5 DOF
- Indirect user interface
- Scanning system (LED)
- 7-slot tray
- Cup holder
- Wheeled cart
- On-board control system



Weight: 25 Kg
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Handy 1 by RehabRobotics, UK

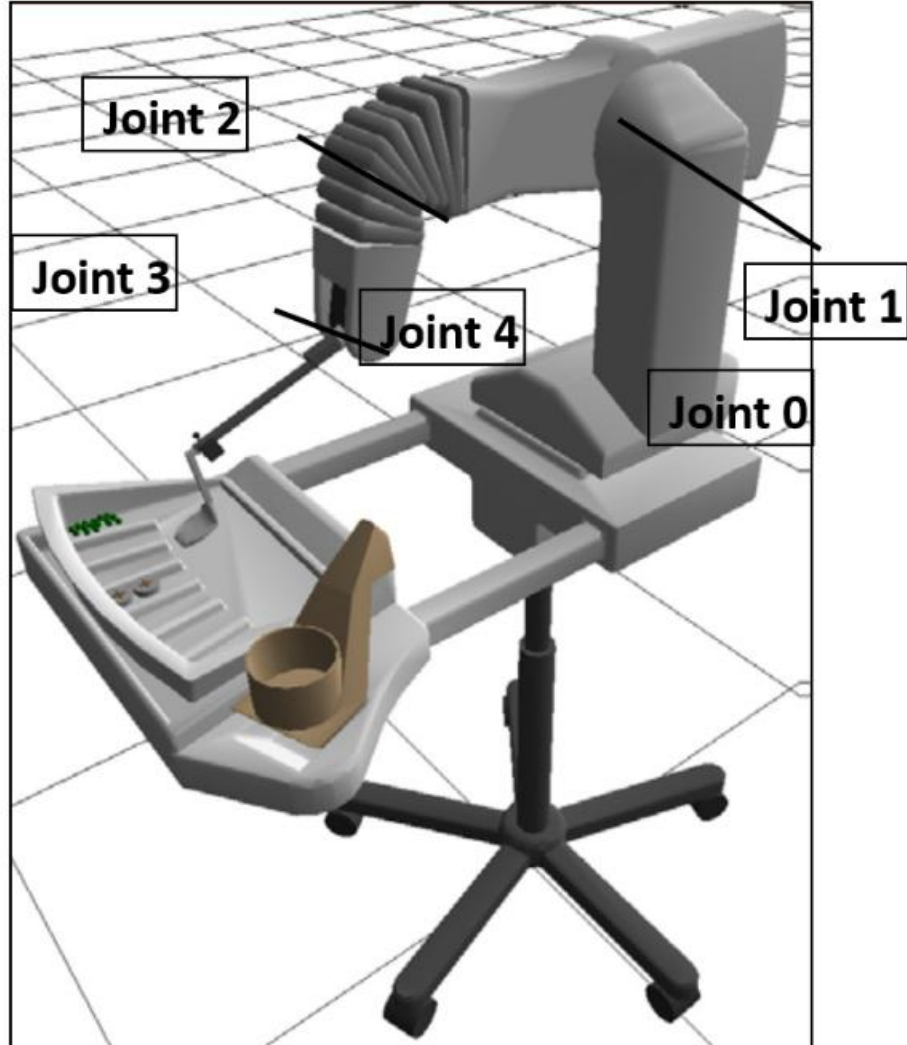
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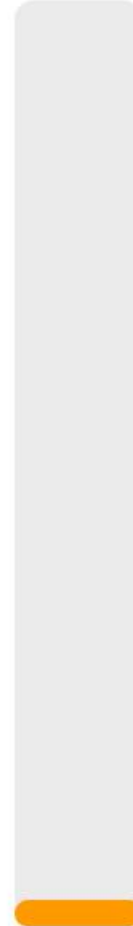
- The user selects the tray slot by scanning
- The spoon reaches the slot
- The spoon moves forward and scoops food
- The spoon moves further, next times (3 positions in total)
- The robot uses $3 \times 7 + 1$ (cup) pre-set positions

How to program Handy1: Is it a structured environment?

✓ 0

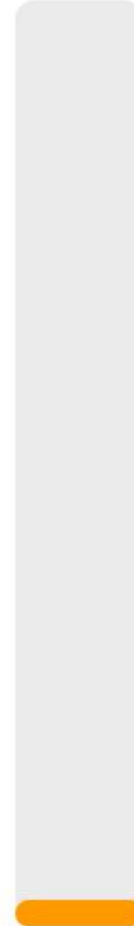


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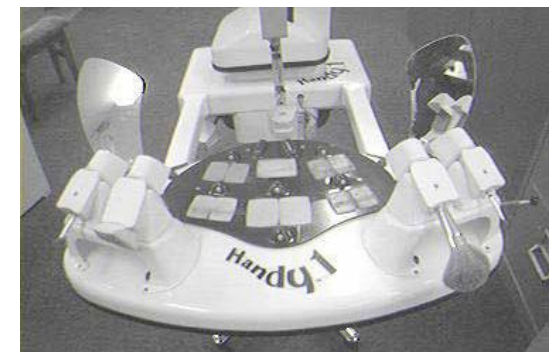
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False

Handy 1 tasks

- **Feeding Set:** LED-based scanning allows the user to select the food slot
- **Cleansing Set:** shaver, toothbrush and cup
- **Make-up Set:** cosmetic products to apply on face



High-tech manipulation aids, for general use

From intro
lecture

Robotic Assistive Technologies

Personal service robots

Assistive Robots (AR)

Socially Assistive Robots (SAR)

Socially Interactive Robots (SIR)

Robotic Assistive Technologies

A sub-class of Assistive Robots (AR), because they are used by individuals with disabilities (while AR are used by any user)

- Robotic prosthetic arms and legs
- Robotic manipulators
- Bionic arms
- Exoskeletons

P. Encarnação, A.M. Cook (2017) *Robotic Assistive Technologies: Principles and Practice*, CRC Press

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High-tech manipulation aids, for general use

Assistive robotic manipulators – ARMs

Differences with industrial robotic manipulators:

- Limited speed and force
- Not in a fixed position
- Human in the control loop

Differences with prostheses:

- Not attached to the human body

Assistive robotic manipulators – ARMs

Definition:

“Any robotic arm that is not body bound with a minimum of 6 DOF and a gripper, which is safe to interact with humans, especially humans with special needs or a disability, and can assist with several activities under the control of the user”

Example of drinking task

Required DOF for Drinking

	Action	Robot Joint Actions	Required DOF
1	Robot holding a glass	Starting point	
2	Robot lifts glass	z-axis movement up and simultaneously movement of gripper down (pitch) to keep glass horizontal	$z + \text{pitch}$
3	Move glass to user	x-axis and y-axis movement toward the user and simultaneous rotation of the gripper on the horizontal plane (yaw) to keep the same gripper orientation	$x, y + \text{yaw}$
4	Start drinking	z-axis movement and roll of gripper	$z + \text{roll}$

Target users of ARMs

Anyone who has no or limited arm or hand function

According to the ICF:

- People with amyotrophic lateral sclerosis (ALS)
- People with muscular dystrophy (MD)
- People with spinal muscular atrophy (SMA)
- People with spinal cord injuries (SCIs)
- Stroke survivors
- People with strong tremor/spasticity

Target users of ARMs

Elderly people constitute another group of potential users of robots that assist in manipulation. Functional decline may create the need for a robot able to retrieve objects for its user.

Individual factors affecting **acceptance** of healthcare robots by older people include the following:

- **Age:** Older people are less likely to imagine living daily with robots. However, acceptance of robots as a way of gaining independence increases in the oldest age range;
- **Needs:** Acceptance of robots by older people increases when there is a perceived need for the device. This is in line with the fact that older people tend to use products because of their functional aspects, in contrast to younger people who often use products because of stylistic considerations regardless of functionality
- **Gender:** Males value more technical characteristics, and females value the emotional aspects of robot interaction
- The more **experience** users have with technology and robots, the greater their level of acceptance
- There is some evidence suggesting that acceptance of robots increases with **cognitive ability** and **level of education**.

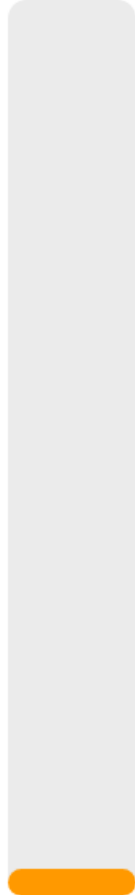
Target users of ARMs

- Strong need for independence
- Good cognitive skills for path planning
- Good cognitive skills for learning the control menu
- Abilities for controlling ARM movements
- Vision
- Powered-wheelchair users, so as to provide battery power supply to the ARM

Tom is 40-year old, had a car accident recently, leaving him quadriplegic. He leaves alone, he is well-educated and has a rewarding professional career. Is Tom a potential ARM user?

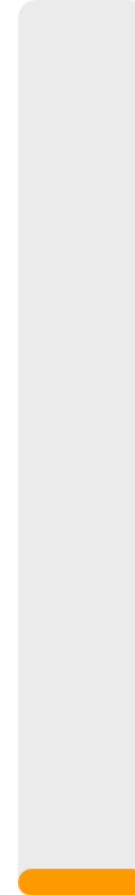


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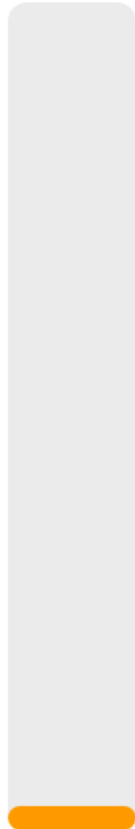


False

Rose is 83 year old and recently suffered from stroke, leaving her emiplegic. She started having mild cognitive issues. She leaves with her husband and has little familiarity with technology. Is Rose a potential ARM user?

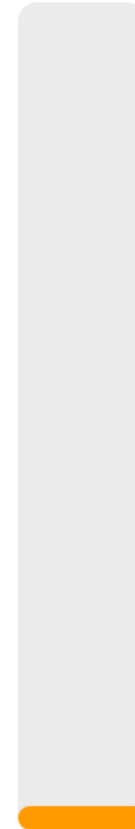


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True

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False

ARM tasks

- Grasping and handling objects
- ADLs like:
 - Bathing and showering
 - Dressing
 - Self-feeding and –drinking
 - Personal hygiene and care (including combing hair, brushing teeth)
 - Toilet hygiene (getting to the toilet, cleaning oneself, getting back up)
 - Doing housework
 - Preparing meals
 - Shopping
 - Using telephone or other form of communication

ARM control

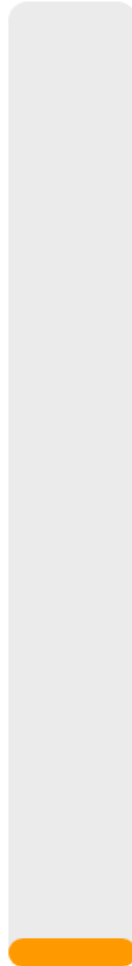
- Let's consider a joystick as an interface → 2 control signals
 - see HTIs for adaptation and access to the joystick
- Control of n DOFs, plus a gripper

How to control the ARM?

How to control an ARM with n DOF, with 2 control signals? In which space?

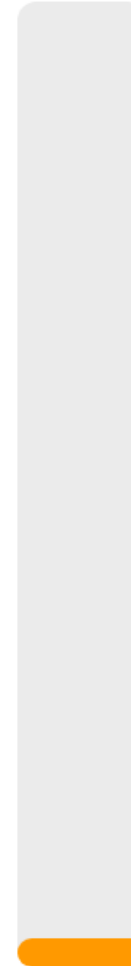
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Joint space

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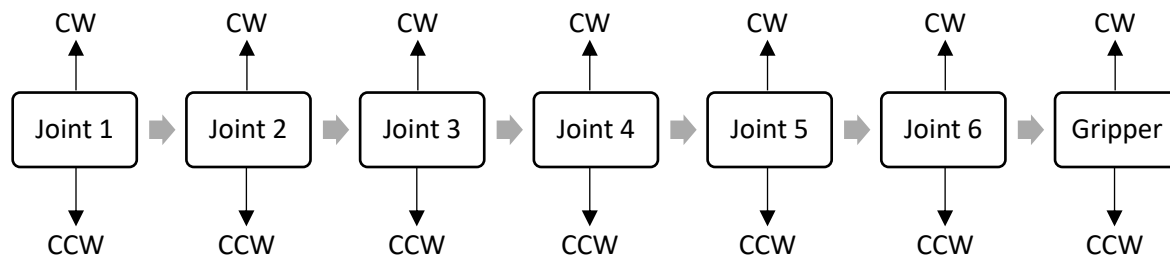


Cartesian space

ARM control: joint vs. Cartesian

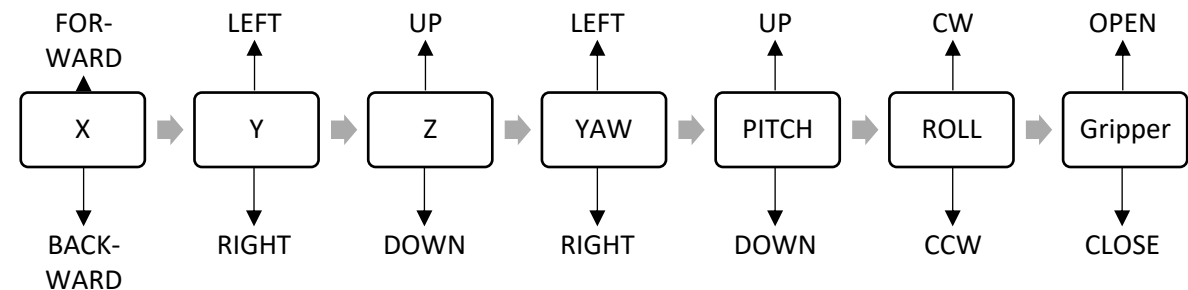
Joint control

- DOFs are controlled one at a time
- Some training is needed for mapping joint rotations to Cartesian desired position
 - **Forward kinematics**



Cartesian control

- Control of (x,y,z) end effector position
 - **Inverse kinematics**, done by the ARM controller
- Gripper orientation can be kept unchanged during movement



Levels of Autonomy

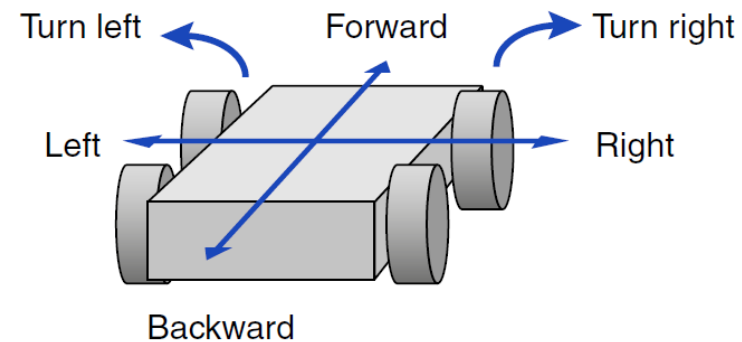
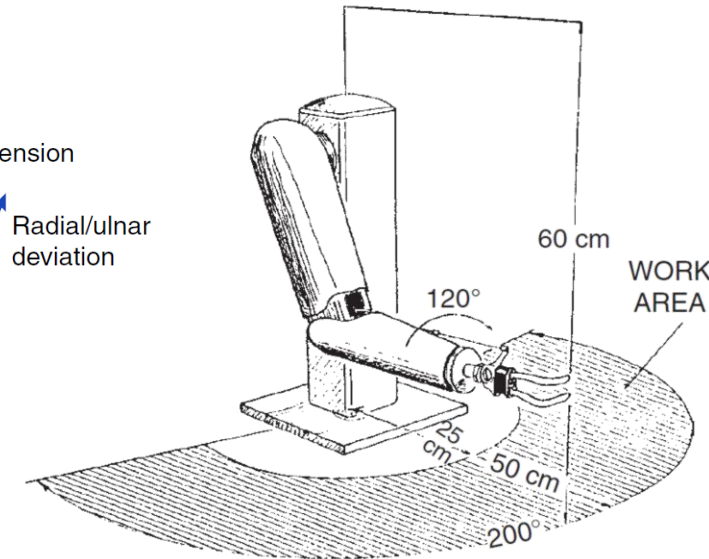
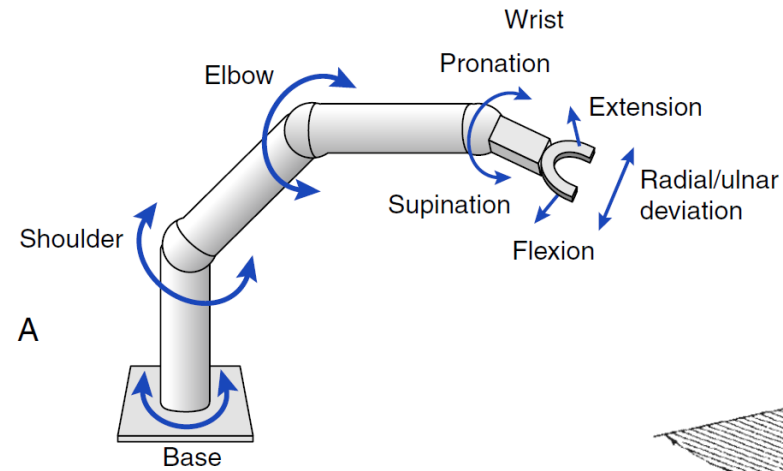
- **Teleoperated:** the user has direct control over the robot movements throughout the task. Multiple controls are then necessary to make the robot move and orient itself in the three-dimensional space.
- **Semi-autonomy levels:** the user merely needs to hit, or press and hold, a switch to replay a prestored movement.
The rationale is to provide users as much control over the task as they want. While in some cases the user may rely on the robot to perform the task all by itself (e.g., reach out for an object on the floor), there are other cases when the user would like to still keep some of the control to influence how the task is performed (e.g., drinking from a glass).
- **Fully autonomous:** the robot can accept high-level commands that define a task to be completed (e.g., take a glass from the shelf, fill it with water, and bring it to the user). The robot will perform whatever subtasks are necessary and make necessary decisions (e.g., pouring water into the glass and determining the glass is full) without requesting any human intervention. Higher levels of robot autonomy require advanced sensing and decision capabilities. The robot needs to be able to sense its environment, which is variable depending on the location and time, and decide on the best approach to carry out a given task.

BOX 15.1 Sheridan's Levels of Robotic Autonomy (Parasuraman et al., 2000)

1. Computer offers no assistance; human does it all.
2. Computer offers a complete set of action alternatives.
3. Computer narrows the selection down to a few choices.
4. Computer suggests a single action.
5. Computer executes that action if human approves.
6. Computer allows the human limited time to veto before automatic execution.
7. Computer executes automatically then necessarily informs the human.
8. Computer informs human after automatic execution only if human asks.
9. Computer informs human after automatic execution only if it decides to.
10. Computer decides everything and acts autonomously, ignoring the human.

ARMs categories

- Robotic workstations
 - Table-mounted or
 - Attached to overhead tracks
- Wheelchair-mounted
- Mobile robots

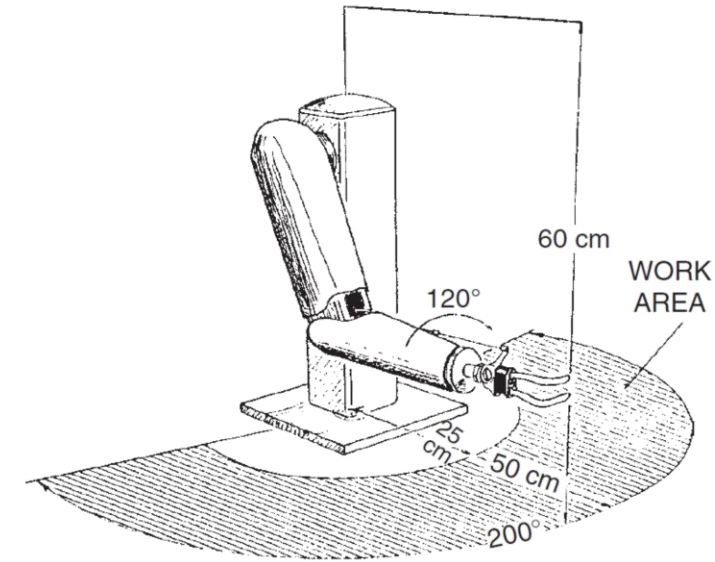


Robotic workstations

APL RAWT

DeVAR/ProVAR

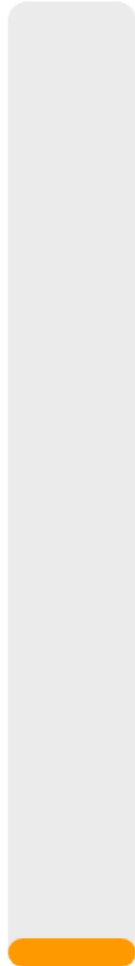
RAID



Robotic workstations - Is the environment structured?

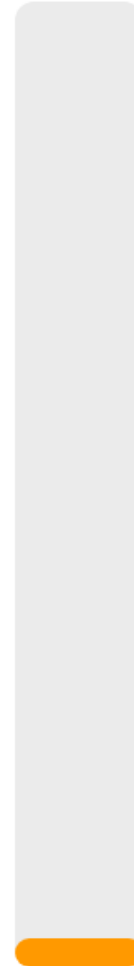


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Yes

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No

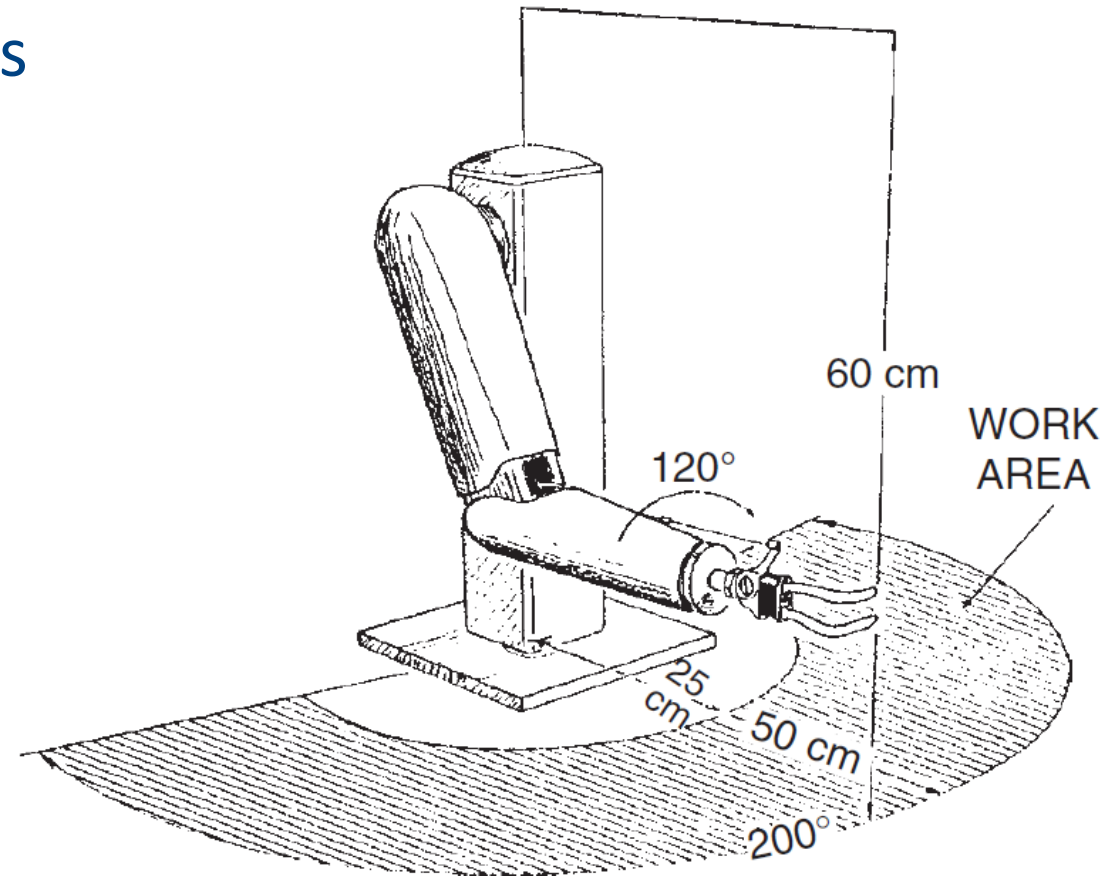
Robotic Workstations

- Workstation: an area dedicated to the performance of a specific job or activity. Examples:
 - design (e.g., a computer workstation for engineering students),
 - reading (e.g., a library-based workstation),
 - clerical tasks (e.g., a workstation for word processing, telephone answering, and manipulation of files).
- These activities involve manipulation of papers, books, and devices.
- Because the workstation is fixed in one location, the design of the robotic system can focus on manipulation of objects in a **structured** environment, without having to deal with uncertainties (e.g., obstacles that may get in the way) characteristic of unstructured environments.

APL RAWT

(Applied Physics Laboratory Robot Arm WorkTable)

- Built from an upper limb prosthesis
- 4 DOF:
 1. Shoulder flexion/extension
 2. Elbow flexion/extension
 3. Wrist pronation/supination
 4. Hand opening/closing
- 2 DOF in the base:
 1. rotation
 2. forward/backward



DeVar/ProVar

VA R&D Rehabilitation Center, Palo Alto, CA

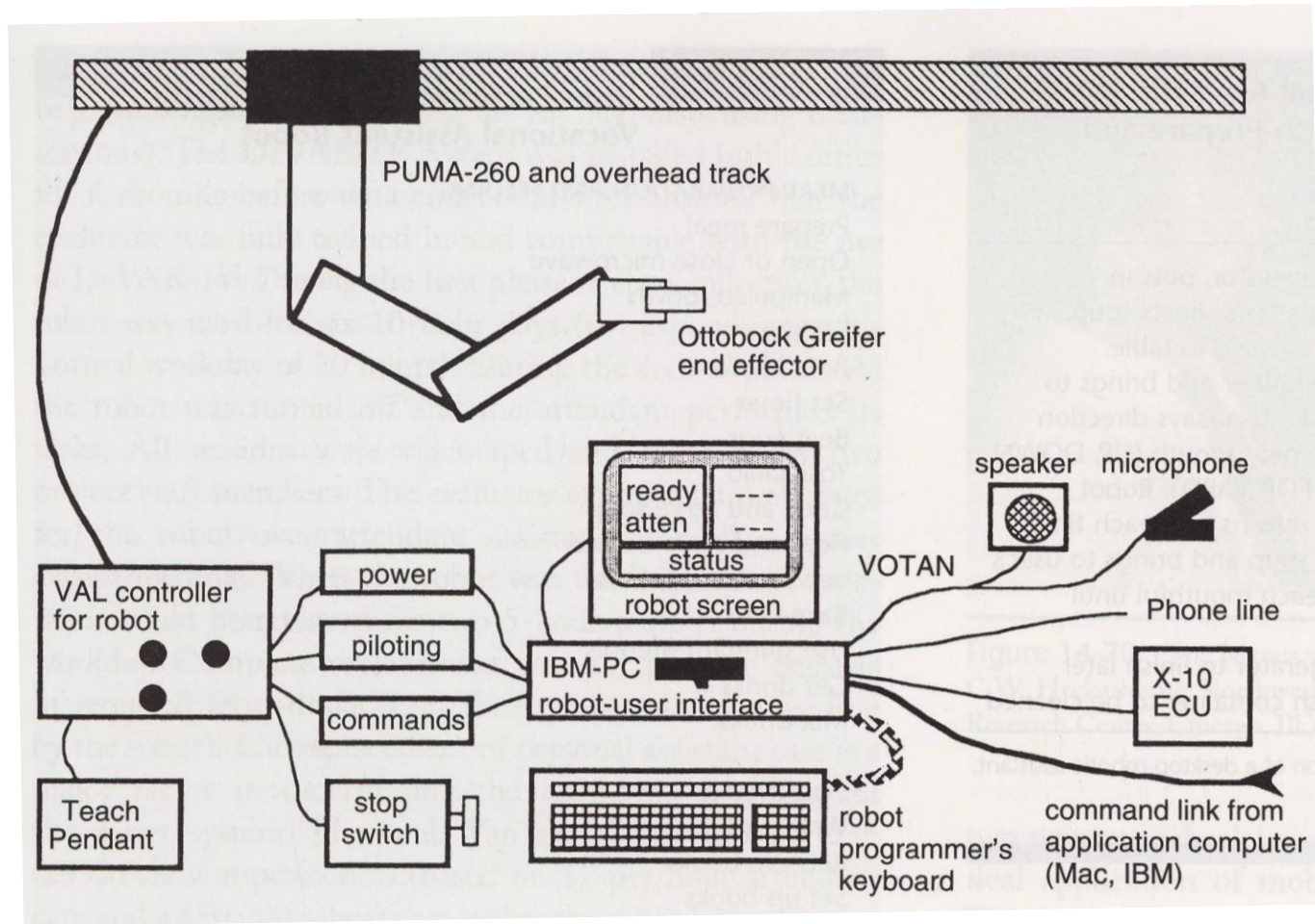
- A PUMA robot used in a workstation
- Ottobock hand prosthesis as end effector



First prototype: 1982

DeVar control

VA R&D Rehabilitation Center, Palo Alto, CA



First prototype: 1982

MASTER/RAID

- The user was able to control a robot mounted on a horizontal rail that ran in front of a metallic structure with shelves
- A vertical axis enabled the robot to access shelves at different heights, and a gripper mechanism provided a means to retrieve objects from the shelves
- These objects could then be placed on the user's desk



First prototype: 1993



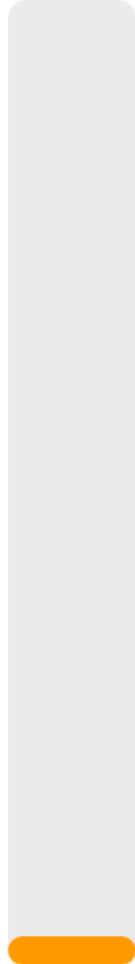
Wheelchair-mounted robotic arms

Manus/iARM
JACO

Wheelchair-mounted robotic arms - Is the environment structured?

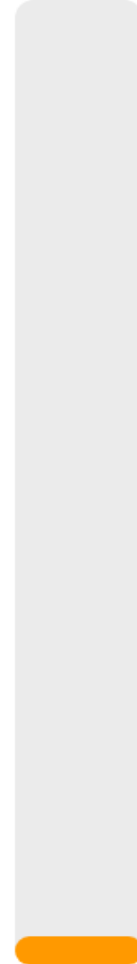
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Yes

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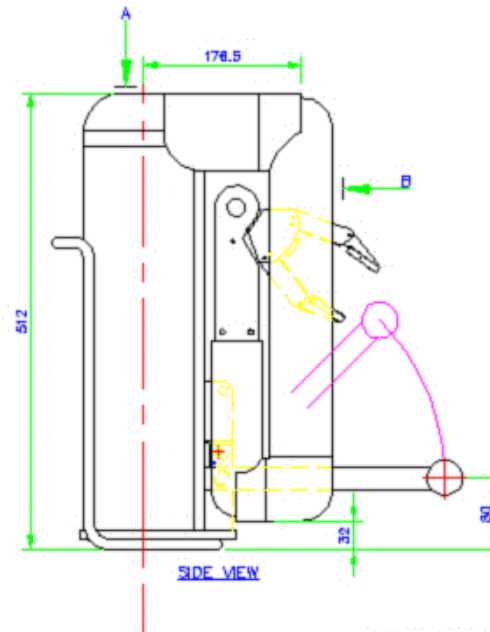
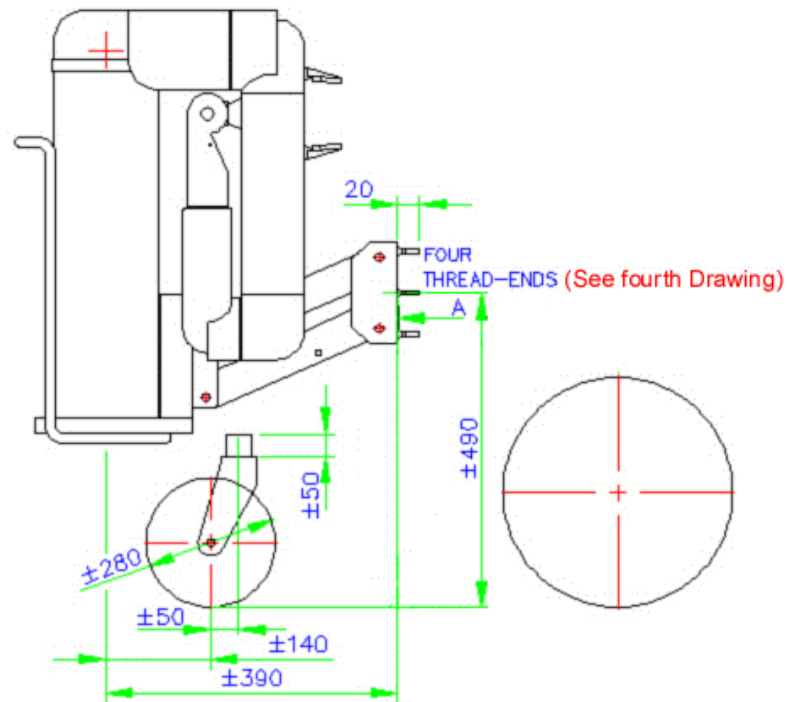
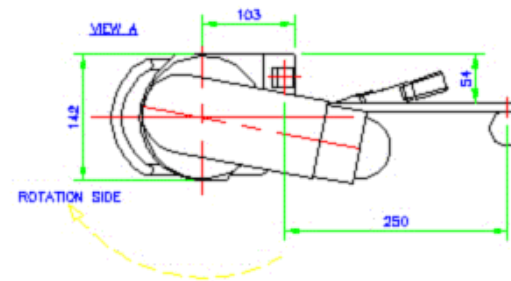
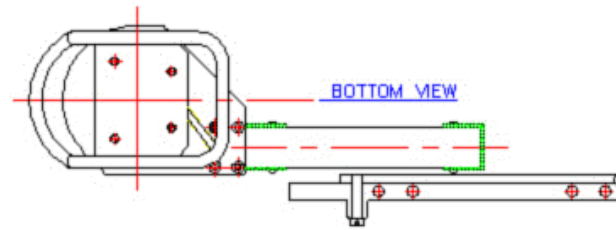
No

Manus / iARM by Exact Dynamics, The Netherlands

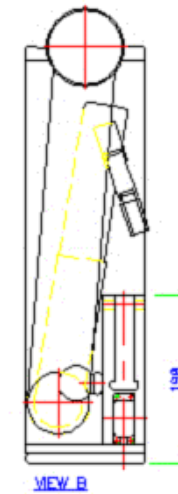
- 8 DOF
- Weight: 9 Kg
- Payload: 1.5 Kg
when fully extended
- Gripping force: 20 N
- Backdrivable belt drive
- Uses wheelchair battery as
source of power



Manus / iARM geometry and kinematics



REMARK: LEFT SIDE MOUNTING



Manus / iARM control

- It can be controlled using
 - direct selection through a keypad or a joystick or
 - indirect selection (scanning) through a single switch
- It allows Cartesian control, enabling the user to move the robot gripper up/down, left/right, and forward/backward while the robot automatically adjusts each joint to keep a cup or plate level during movement.
- It stores particular positions that are frequently used. They can be recalled with one selection from a menu.



JACO by Kinova, Canada

- 7 DOF
 - 3-DOF wrist: pronation/supination, flexion/extension, radial/ulnar deviation
- Equipped with a two-finger or a three-finger hand
 - 1-DOF hand: open and close
- The end effector can be moved in 3D space
 - up/down
 - left/right
 - forward/backward
- Maximum reach 900 mm
- Weight: 5.2 Kg
- Payload: 1.6 Kg
- Lighter 6-DOF version available (4.4 kg)



A hand is holding a black and white robotic arm, identified as the JACO arm, against a solid blue background. The arm is positioned diagonally across the frame. Overlaid on the image in large, white, bold, sans-serif capital letters with a black outline is the text "MEET JACO".

MEET JACO

https://www.youtube.com/watch?v=_Udif0YGR-Q&t=137s

SPECIFICATIONS



Weight	5.2 kg / 11.5 lbs
Payload (full-range continuous)	1.3 kg / 2.9 lbs
Payload (mid-range continuous)	1.6 kg / 3.5 lbs
Maximum reach	900 mm / 2.95 ft
Materials	Carbon fiber and aluminum
Maximum speed	20 cm/s
Power supply voltage	18 to 29 VDC
Average power	25 W (5 W on standby mode)
Peak power	100 W
Ingress protection	IPX2
Operating temperature	-10 °C to 40 °C / 14 °F to 104 °F



Reach us 

<https://assistive.kinovarobotics.com/product/jaco-robotic-arm>



JACO control

- 3 control modes, accessed by 2 push buttons
 1. **Translation mode:** the user can move the robot's hand in 3D space with the orientation of the hand fixed
 2. **Wrist mode:** the user can modify the orientation of the hand, but the hand is kept centered at the same point in space.
 3. **Finger mode:** the user can grasp and release the hand, using either 2 or 3 fingers

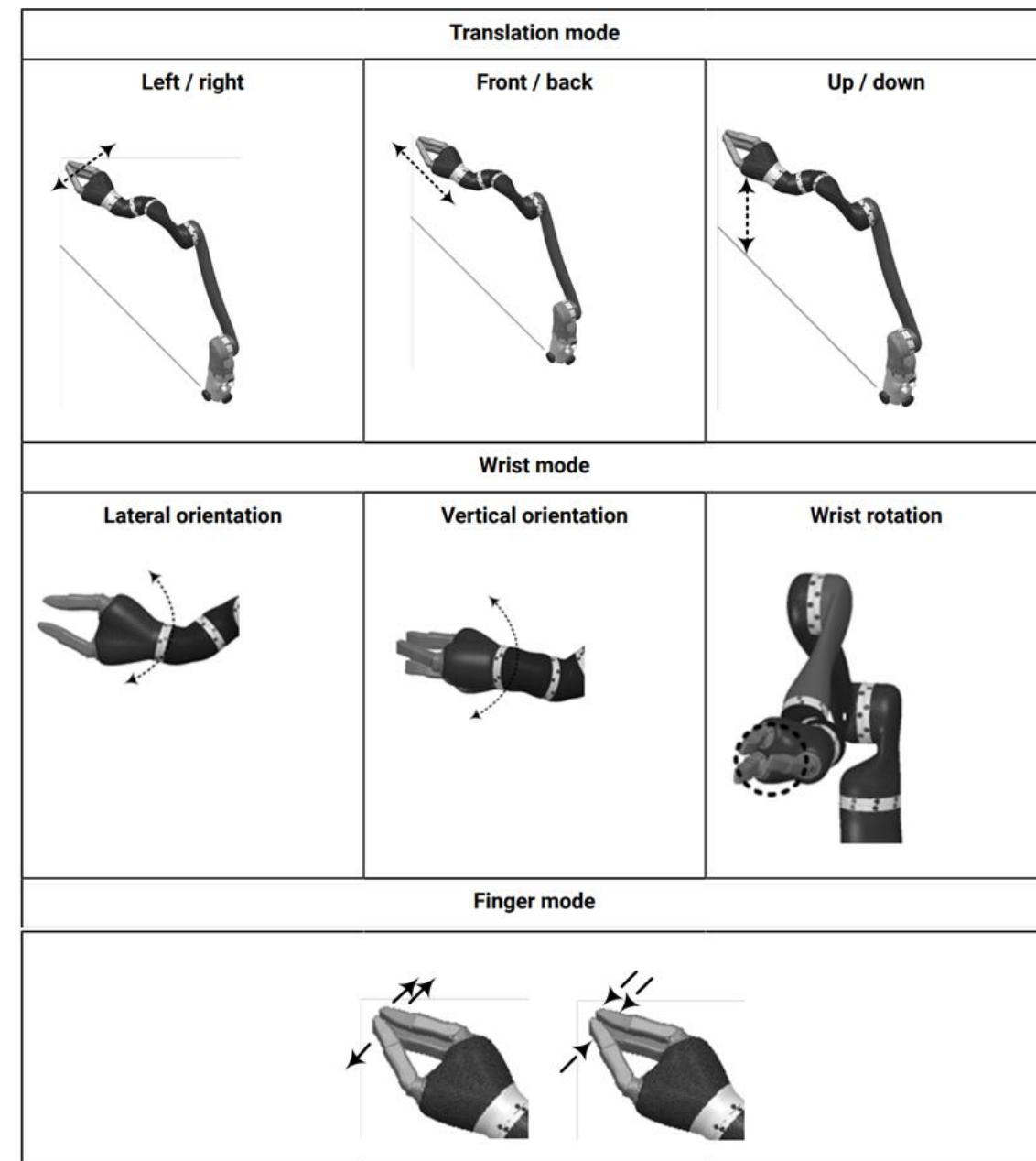
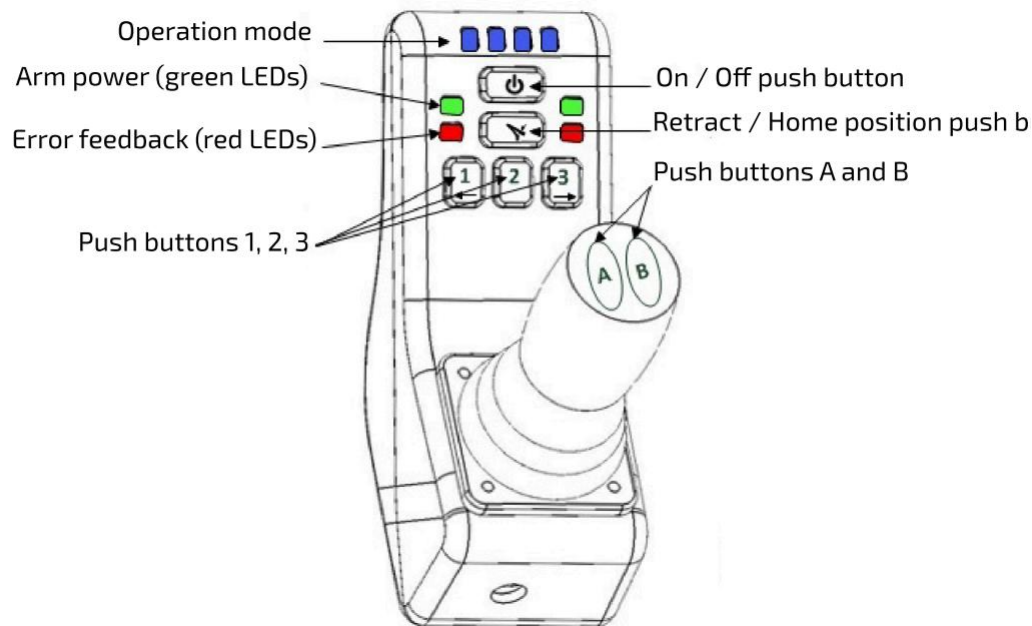


Table 4: Three-axis mode joystick controls

JACO control

- 3-axis joystick, mounted on the wheelchair armrest
 1. shaft forward or backward
 2. shaft right or left
 3. handle turned clockwise and counter clockwise



Joystick movement	Robot movement	Availability
Translation mode		
Incline FRONT	Gripper moves forward	4 / 6 / 6S / 7S DoF
Incline BACK	Gripper moves backward	4 / 6 / 6S / 7S DoF
Incline LEFT	Gripper moves left	4 / 6 / 6S / 7S DoF
Incline RIGHT	Gripper moves right	4 / 6 / 6S / 7S DoF
Rotate stick CLOCKWISE	Gripper moves up	4 / 6 / 6S / 7S DoF
Rotate stick COUNTERCLOCKWISE	Gripper moves down	4 / 6 / 6S / 7S DoF
Wrist mode		
Incline FRONT	Vertical orientation – top side	6 / 6S / 7S DoF
Incline BACK	Vertical orientation – bottom side	6 / 6S / 7S DoF
Incline LEFT	Lateral orientation – thumb side	6 / 6S / 7S DoF
Incline RIGHT	Lateral orientation – index side	6 / 6S / 7S DoF
Rotate stick CLOCKWISE	Wrist rotation clockwise	4 / 6 / 6S / 7S DoF
Rotate stick COUNTERCLOCKWISE	Wrist rotation counterclockwise	4 / 6 / 6S / 7S DoF
Finger mode		
Incline LEFT	Close Fingers (3-finger mode)	4 / 6 / 6S / 7S DoF
Incline RIGHT	Open Fingers (3-finger mode)	4 / 6 / 6S / 7S DoF

<https://assistive.kinovarobotics.com/uploads/EN-UG-007-Jaco-Assistive-robot-user-guide-r06.1.pdf>

JACO – iARM comparison

Characteristics of the Commercially Available JACO and iARM

Characteristic	iARM	JACO
Manufacturer	Exact Dynamics	Kinova Technology
Weight	9 kg	5 kg
Weight limit	1.5 kg	1.5 kg (45 cm), 1.0 kg (90 cm)
Reach	90 + 20 cm (lift unit)	90 cm
Maximum speed	15 cm/s	15 cm/s
Degrees of freedom (DOF)	7, including gripper; 8 with lift unit	7, including gripper
Hand	2 fingers	3 fingers
Finger force	20 N	7 N
Control possibilities	Several power wheelchair interfaces, keypad, joystick (digital or analog), single switches, game pad, computer	Several power wheelchair interfaces, mini-joystick, three-axis joystick, button controls, game pad, computer
Power	24-V DC/3 A (maximum)	24-V DC/1.5 A

Source: Adapted from Chung, C. S., and Cooper, R. A. 2012. “Literature review of wheelchair-mounted robotic manipulation: user interface and end-user evaluation.” Proceedings of the 12th Annual RESNA Conference, Baltimore.



Mobile robots

MOVAID

Care-o-Bot

Mobile robots as assistive devices

- Wheeled mobile robots equipped with an arm and a simple tray to carry objects or capable of controlling the environment
- Common features:
 - monitoring the user, including video-monitoring by a caregiver or health care provider
 - helping the user to accomplish activities of daily living, either by providing
 - cognitive support (e.g., warning the person that it is time to take a pill or prompting the person through the steps of cooking a meal)
 - physical support (e.g., retrieving a glass of water)

MOVAID

(Mobility and Activity Assistance Systems for the Disabled)

- 2-DOF mobile base
- 8-DOF robotic arm
- 2-DOF 3-finger hand
- 2-camera vision system
- Laser-based localization
- US obstacle avoidance
- Tray for object transportation
- Docking system
 - Fixed workstations in the home environment



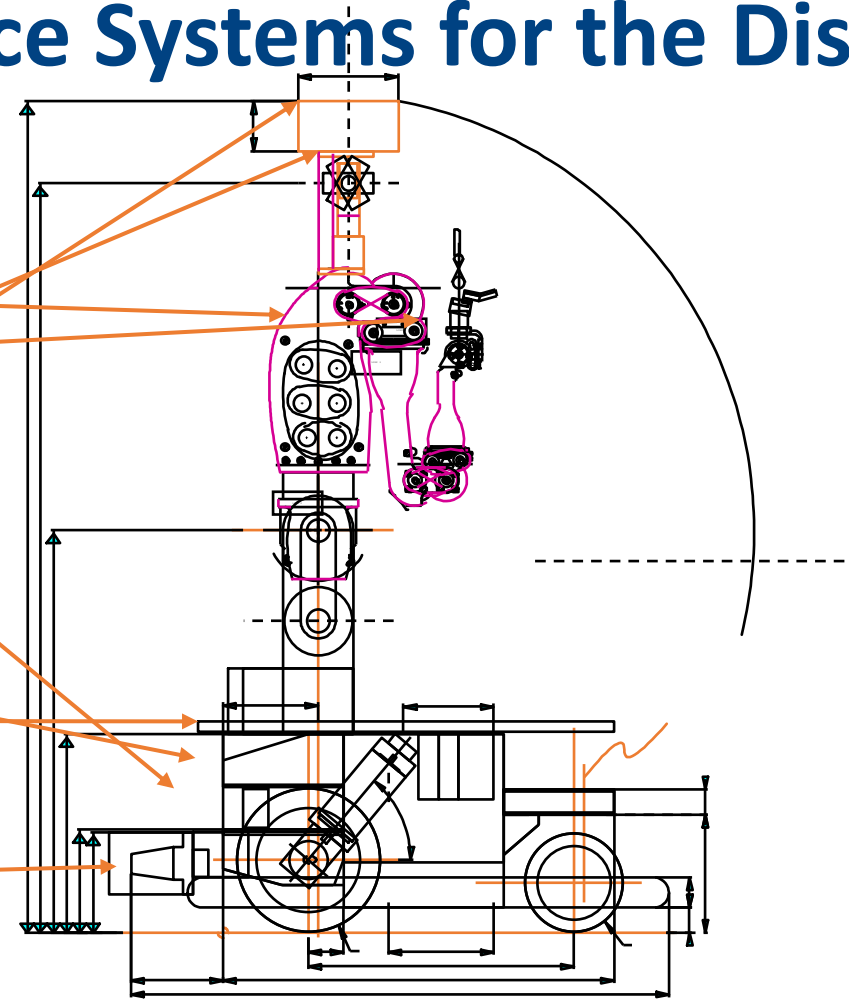
EC TIDE project, Scuola Superiore Sant'Anna, Pisa, 1994-1997

Dario P., Guglielmelli E., Laschi C., Teti G.(1999) "MOVAID: A personal robot in everyday life of disabled and elderly people" *Technology and Disability* 10(2):77–93.

MOVAID

(Mobility and Activity Assistance Systems for the Disabled)

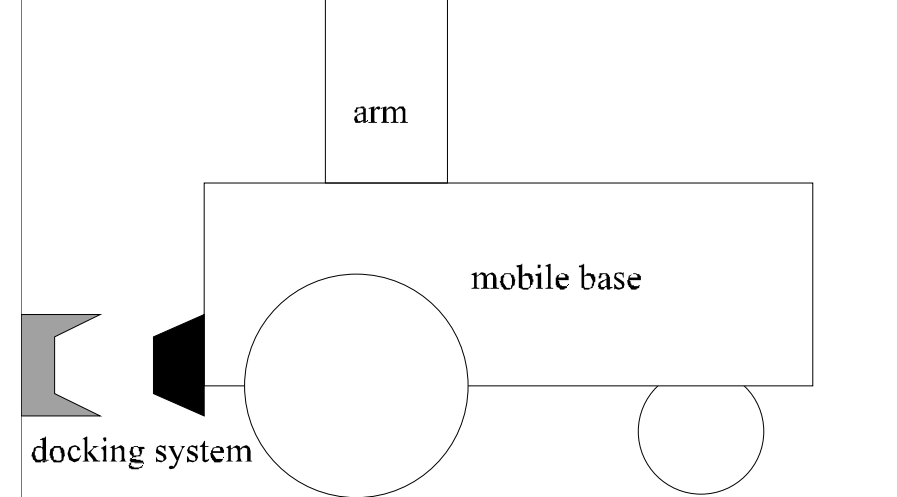
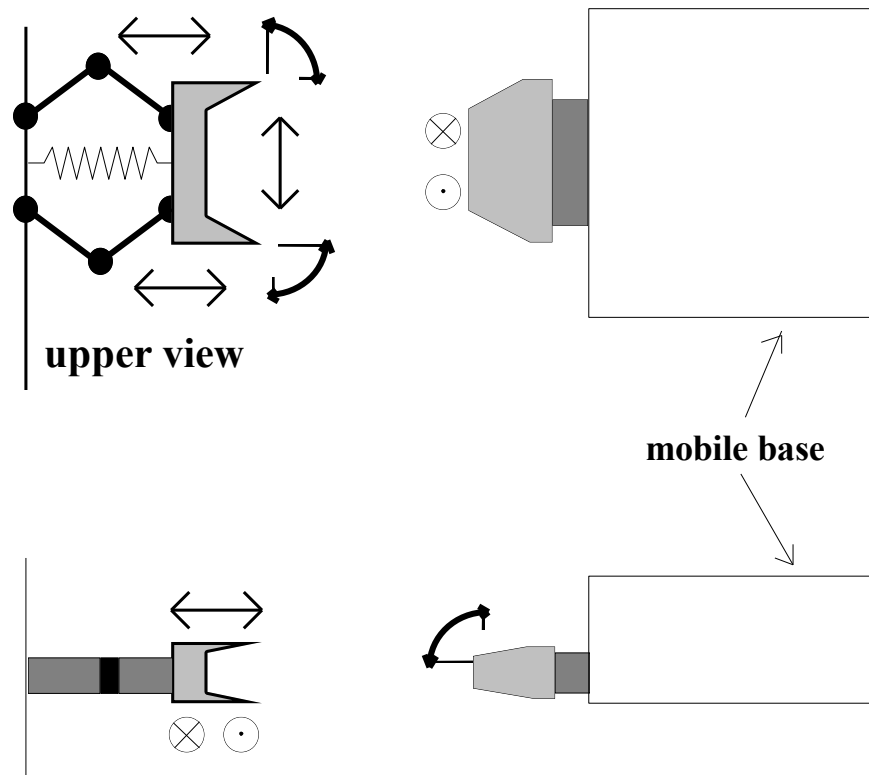
- 2-DOF mobile base
- 8-DOF robotic arm
- 2-DOF 3-finger hand
- 2-camera vision system
- Laser-based localization
- US obstacle avoidance
- Tray for object transportation
- Docking system
 - Fixed workstations in the home environment



EC TIDE project, Scuola Superiore Sant'Anna, Pisa, 1994-1997

MOVAID docking

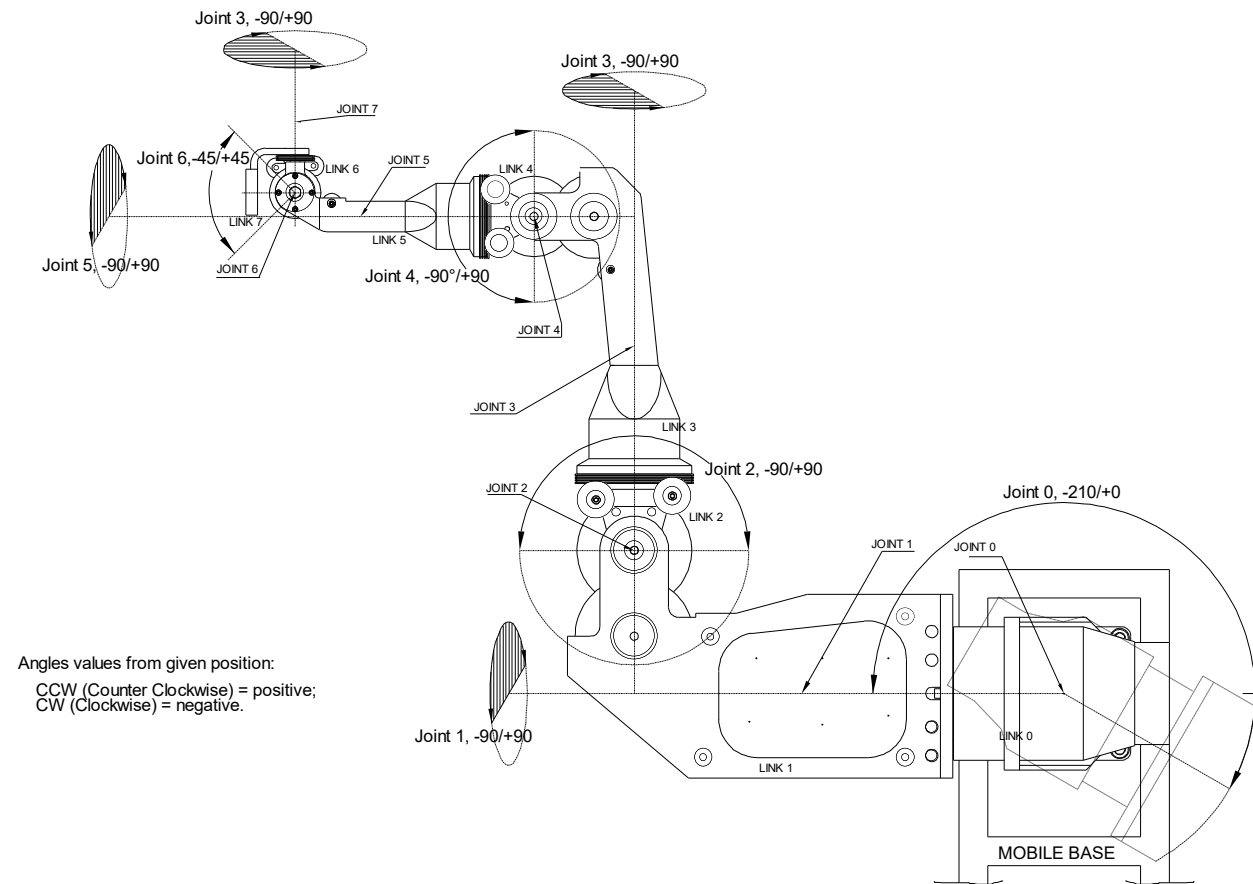
When docked, the robot operates in a structured environment, similarly to a robotic workstation



side view

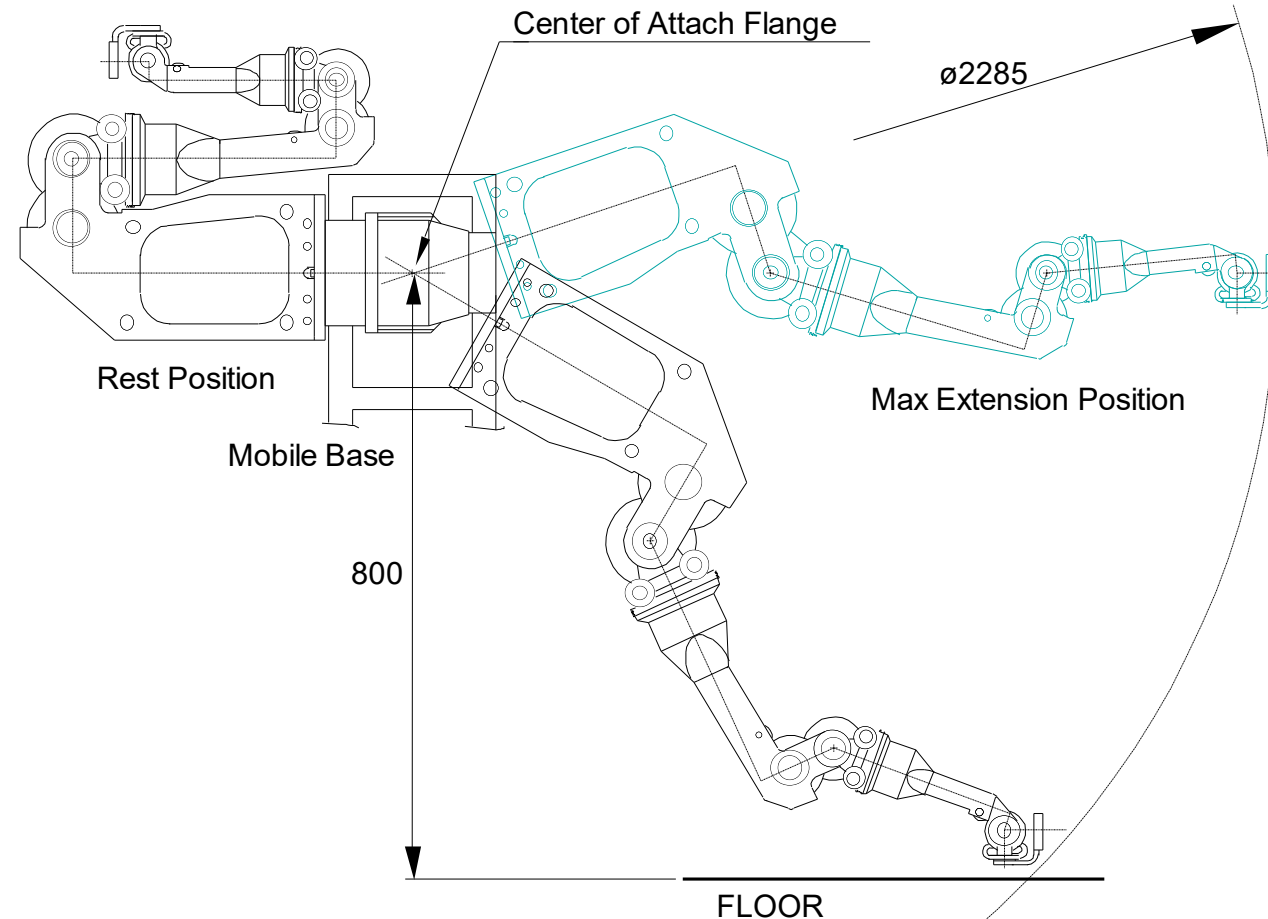
Dexter arm by Scientia Machinale, Italy

- 8 DOF
- Maximum reach: 1500 mm
- Repeatability: + 1mm
- Maximum speed: 0.2 m/s
- Weight: 40 Kg
- Payload: 2 Kg
- Cable-driven: reduced size, since motors are outside the arm



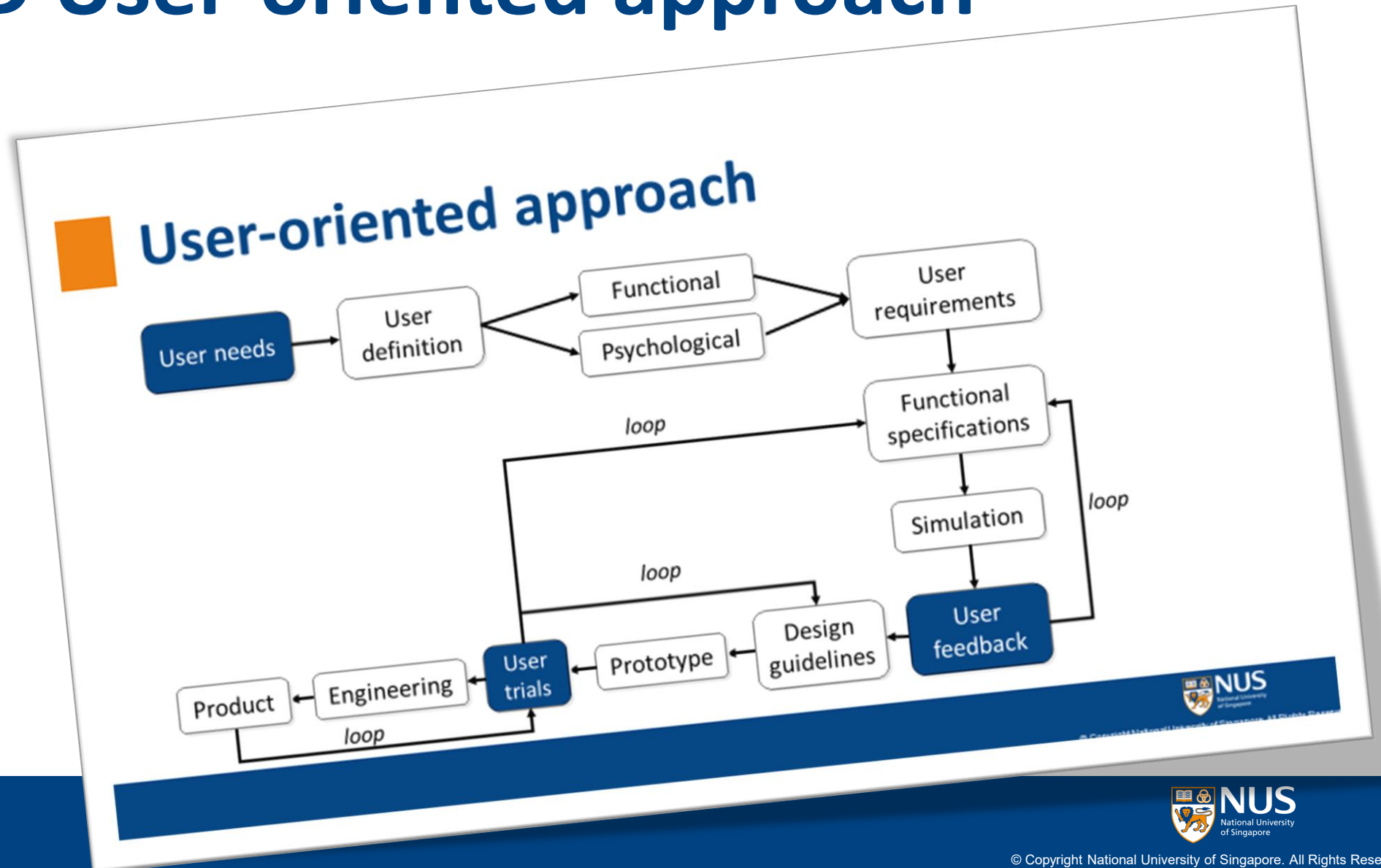
Dexter arm by Scientia Machinale, Italy

- Workspace: 1200 mm x 350°
- Foldable structure:
 - 400 mm x 400 mm x 950 mm
- Sagittal plane workspace is a circle of diameter 2285 mm
- The arm can reach an object
 - on a 400-mm wide shelf at 1500 mm height
 - on the floor

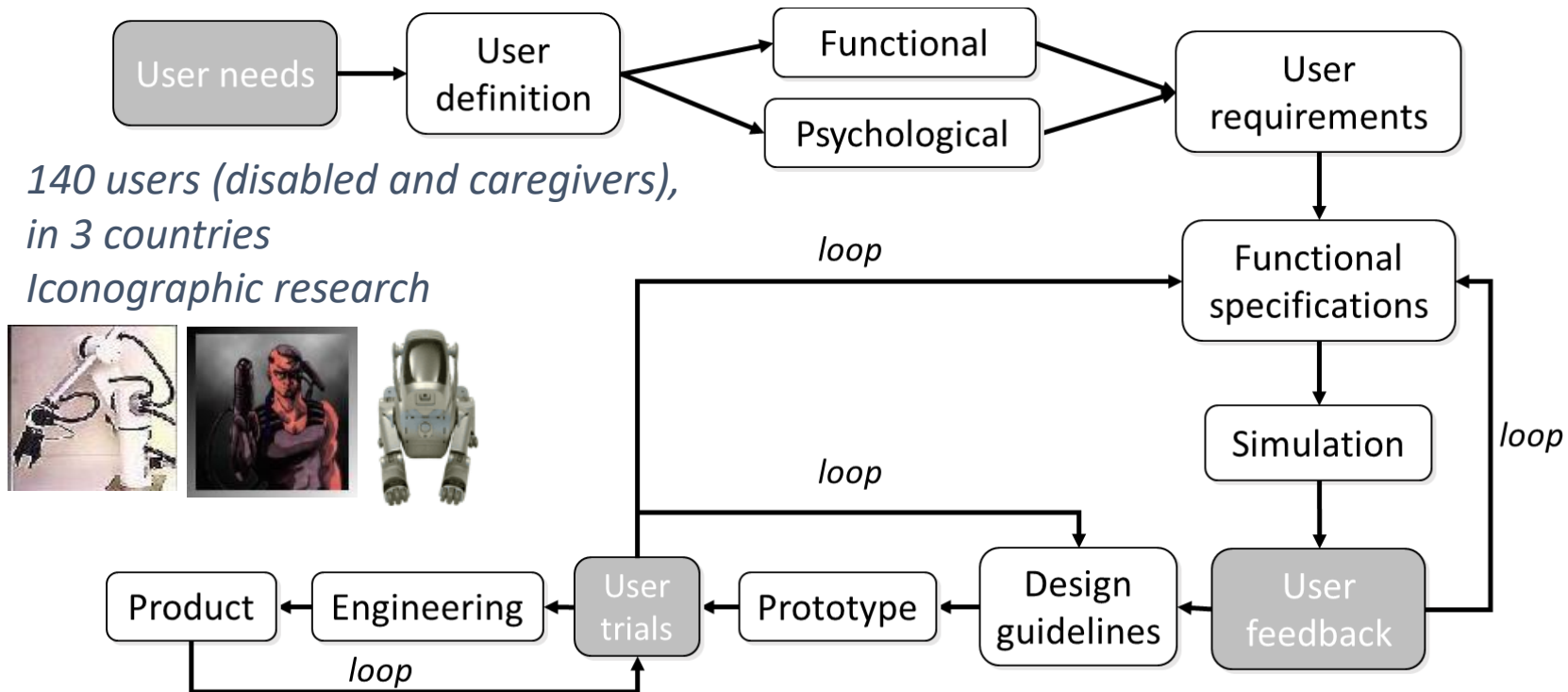


MOVAID User-oriented approach

From intro
lecture

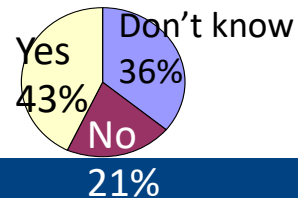


MOVAID User-oriented approach



70 users, 7 validation sites (hospitals,
rehabilitation centres) in 3 countries:
Italy, France, Switzerland

Would you like an assistive robot?



3 demo tasks

- Dish warm-up
- Kitchen countertop cleaning
- Bed sheet removal

Semi-autonomy

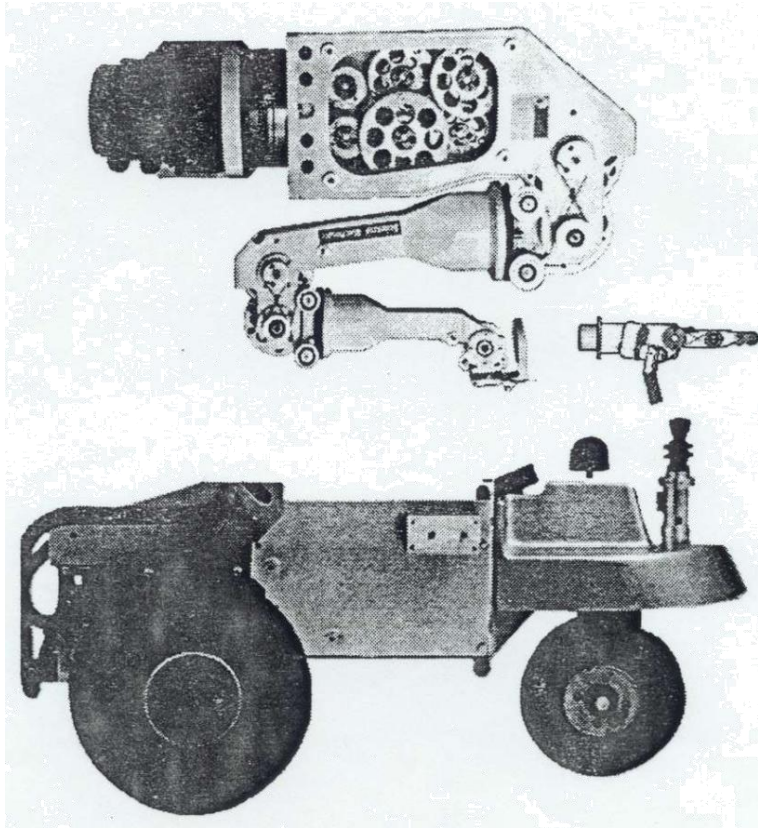


Desirable characteristics of a personal robot according to users analysis

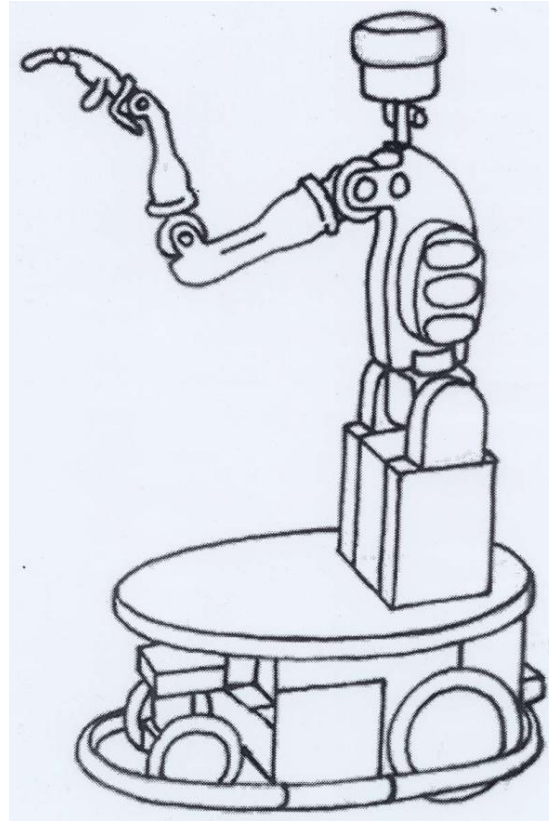
- the robot should look clearly **artificial**, never biomorphic nor anthropomorphic;
- a careful balance should be kept between the technical look and some 'softer' elements suggesting a feeling of humanism and **friendliness**;
- '**caricature**' elements (pet-robot, puppet) should be generally avoided, as they could create a sense of unreliability (should I be helped by a toy?);
- the robot should 'sleep' in a **reassuring position** and, when waking up, it should **warn** and start with a **gentle movement**;
- the robot should never move in the darkness, or should **warn** with a light before moving;
- the arm should possibly offer objects from a **lower position**, avoiding coming toward the user with a 'threatening' attitude.

By industrial designers

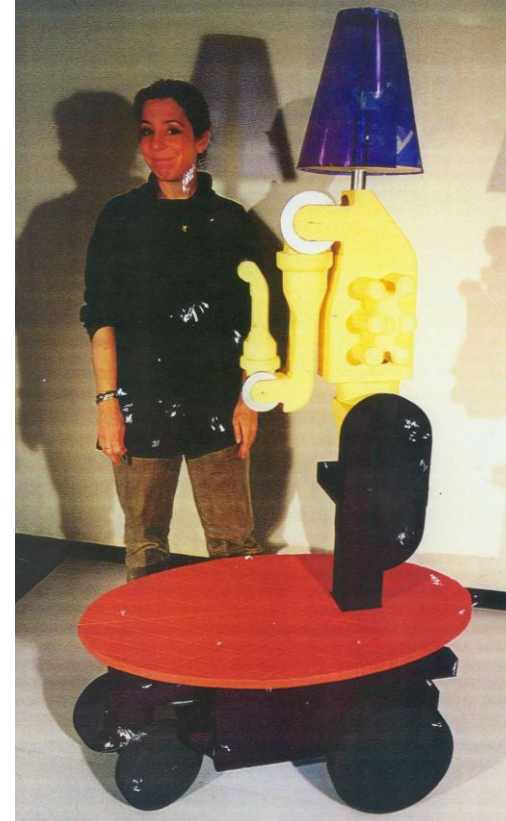
MOVAID industrial design



Starting components



Aesthetic design



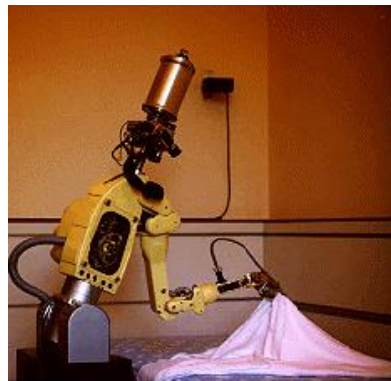
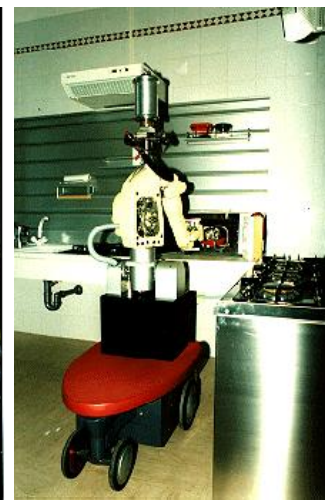
Wooden mock-up



Prototype

MOVAID tasks

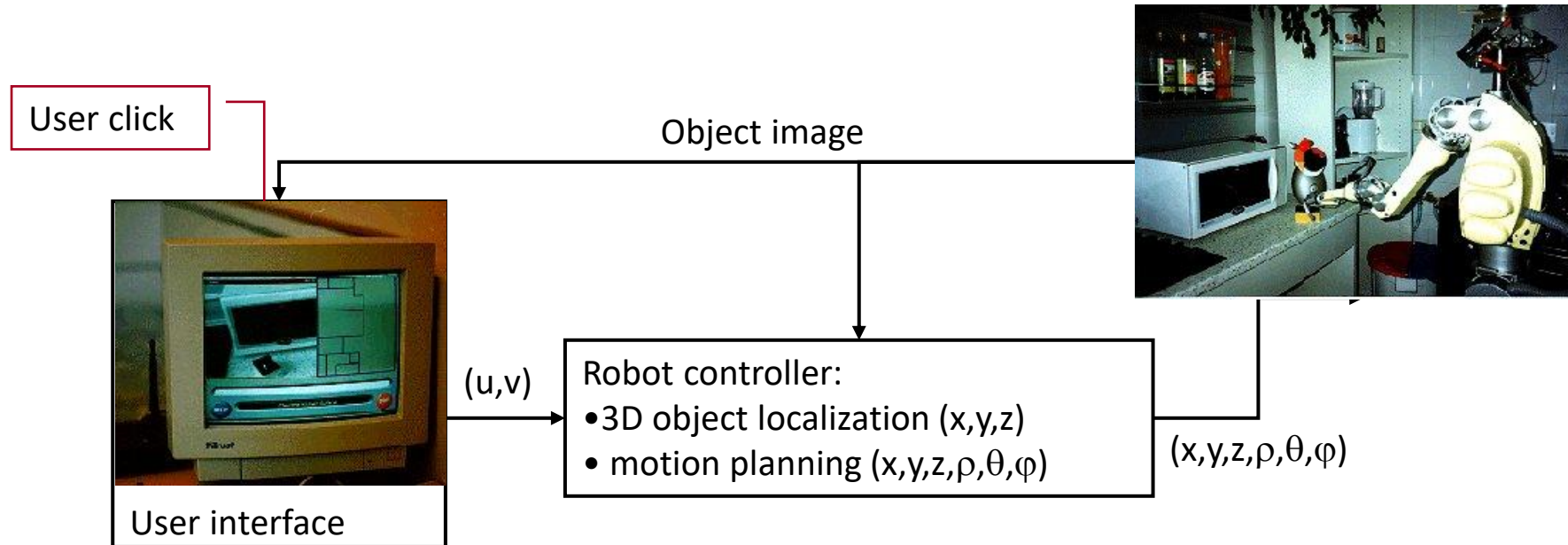
1. Dish warm-up
2. Kitchen countertop cleaning
3. Bed sheet removal



MOVAID semi-autonomy

User-robot cooperation

- The user helps the robot in tasks easy for the user and difficult for the robot; e.g., object recognition
- The robot helps the user in manipulation task



Care-o-Bot

- Household robot that assists people in performing their everyday life at home, such as cleaning table and clearing dish washer.
- Not limited to assisting older people



TECHNICAL DATA SHEET



Care-O-bot 4



MODULE	COMPONENT/ PROPERTY	VALUE
FULLY CONFIGURED ROBOT		
	Dimensions (L/W/H):	72/72/158 cm
	Weight:	140 kg
	Power Supply:	Li-Ion battery 48V External charger 48V@10A Internal levels 48V, 24V, 19V (battery voltage can vary 44V-56V)
	Degrees of freedom: Safety:	29 3 safety laser scanners with safety PLC 2-channel Safe-Torque-Off (STO) separate for mobile base and torso 2-channel EM-Stop prepared in all modules 2-channel wireless EM-stop Gigabit Ethernet in all modules 4-6 Intel NUC i5, 256GB, 8GB RAM
	Communication: Computing:	
	Dimensions (L/W/H):	72/72/33 cm
	Weight:	42 kg
	Power Supply:	Li-Ion battery 48 V External charger 48V@10A Internal levels 48V, 24V, 19V
	Communication:	Gigabit Ethernet / Dual-channel router with DDWRT (2,4GHz+5GHz) / Low-level: CANOpen
	Computing Interface:	1x Intel NUC i5, 256GB, 8GB RAM
	MaxSpeed:	7" touchscreen, power button, break-release, USB port, LAN port 1.1 m/s
	Dimensions (L/W/H):	43/80 cm
	Weight:	43,2 kg
	Power Supply:	Power supply from base 48V Internal levels 48V, 24V, 19V
	Communication:	Gigabit-Ethernet
	Computing Interface:	1-3x Intel NUC i5, 256GB, 8GB RAM
	Lighting:	7" touchscreen 58x independent RGB-LEDs in ring
	Torso joint 1-DOF:	Max Joint Vel: 200°/sec Joint Limits: -360° to 360°
	Torso joint 3-DOF:	Max Joint Vel: 200°/sec per axis --> ~ ca. 1 bow/sec Joint Limits: -360° to 360° all axis

Care-o-Bot

- Household robot that assists people in performing their everyday life at home, such as cleaning table and clearing dish washer.
- Not limited to assisting older people



TECHNICAL DATA SHEET



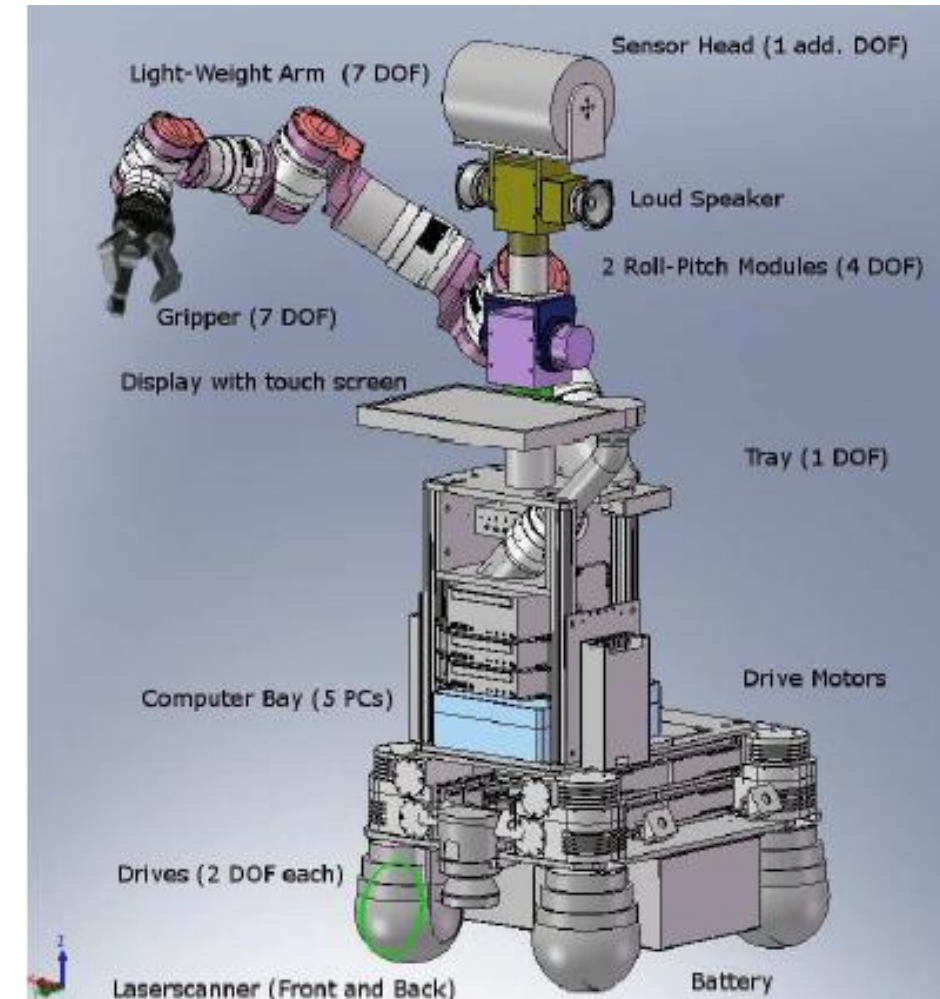
Care-O-bot 4



MODULE	COMPONENT/ PROPERTY	VALUE
ARM AND GRIPPER 	ARM	Reach: ca. 90 cm (from center of first ball) Weight: 13,9 kg Degrees of freedom: 7 Communication: CANOpen Motors: Schunk PowerBalls (PRL100, ERB145, ERB115) Payload: 5kg
	GRIPPER	Dimensions (L/H/W): 17/12,5/11 cm Weight: 0,9 kg Degrees of freedom: 2 Communication: USB Camera: 3D-RGBD camera (range 15-200 cm) Light: 1x 3W Daylight LED, 1x 1W (per channel) RGB-LED Laser: Laserpointer, red, <5mW Other Interfaces: 6x configurable GPIO-ports 1x 20MHz SoC with logic blocks, USB, free programmable 1x USB port for external sensors
HEAD 		Dimensions (D/H): 37/51 cm Weight: 14,7 kg Communication: Gigabit Ethernet Computing: 1x Intel NUC i5, 256GB, 8GB RAM Interface: 15" touchscreen, 2x 20W speaker, 1x mic
	Head joint 1-DOF: Head joint 3-DOF:	Max Joint Vel: 200°/sec Joint Limits: -360° to 360° Max Joint Vel: 200°/sec per axis Joint Limits: -360° to 360° all axis
SENSOR RING 		Dimensions (DH): 35/9cm Weight: 1kg + custom sensors Communication: Gigabit Ethernet Computing: 1x Intel NUC i5, 256GB, 8GB RAM Speed: 200°/sec Angle Limits: -360° to 360°

Care-o-Bot hardware architecture

- 28 DOF
- 4 wheels, independently controlled in orientation and speed, so that the mobile base is omnidirectional
- 600x600x340mm mobile base, with:
 - Battery pack Li-Io (50 V, 60 Ah)
 - 2 laser scanners
 - Navigation PC
- The torso supports sensors, arm and tray. It contains most electronic components and control PCs
- Total height: 770 mm.
- Robotic arm: Schunk LWA3, 7-DOF
- Attachment for diverse end effectors



Care-o-Bot modular design

- The robot can be equipped with one, two or no arms at all.
- The patented spherical joints on the robot's neck and hips are optional.
- If the intended purpose is to serve drinks, one hand can be replaced by a tray, or the mobile base platform can be used on its own as a serving trolley.
- Individual robot platforms can be configured for a wide range of applications: a mobile information center in museums, DIY stores and airports, for collection and delivery services in homes and offices, for security applications or as museum robots at attractions



As a roboticist, how difficult do you see the following tasks:

✓ 0

Robot navigation in an apartment

Robot recognizing common objects

Robot grasping common objects

Robot understanding user's requests

Robot handing over objects to the user

Summary

- Brief recall of target users
- Classification of manipulation aids
 - High-tech vs. low-tech
 - Specialized vs. general use
- Examples of specialized AMRs (e.g., feeders)
- ARMs – generalities on control
 - Fixed workstations
 - Wheelchair-mounted manipulators
 - Mobile robots